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RESEARCH ARTICLE

The Thi Qar Survey Results from the Umma Region (Iraq)[1]

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1. INTRODUCTION

Settlement surveys were carried out in Iraq starting in the late 1930s with the first broad scale archaeological reconnaissance carried out in the framework of the "Diyala Basin Archaeological Project"(Jacobsen 1960, 174–175). The methods developed in this project were then applied to other archaeological surveys in the south of Iraq. First in the "Survey of Central Sumer" carried out in 1953–1954 by Fuad Safar, Vaughn Crawford and Thorkild Jacobsen (1954, 1958; Goetze, 1955), and slightly later in surveys of the Ur and the Lagash regions carried out by Thorkild Jacobsen (1960, 1969). Georges Roux (1960) surveyed the Hammar Lake District in 1952 and McGuire Gibson (1972) the region surrounding the ancient city of Kish (Tell al-Uhaymir) (see Ur 2013 for a comprehensive overview). Most notable and defining for the field of Mesopotamian Archaeology however, were the surveys carried out by Robert McCormick Adams (1981, 1972, 1965) in cooperation with Hans-Jörg Nissen (1972) and Henry Wright (1981). Adams (1981) surveyed a third of the entire alluvial plain under fairly difficult conditions over a 21-month period, spread over two decades between 1956–1975. Most of the areas he surveyed had not previously been mapped, making navigating the terrain really challenging. In addition, the alluvial plain did not allow for systematic sampling, as seasonally filled depressions, marshes, lakes, agricultural fields with webs of irrigation canals, and sand dune belts impeded moving along predefined transects (Adams 1981, 38).

Certain areas were totally inaccessible, such as the eastern half of Thi Qar province, which was at the time when Adams (1981, 36) carried out his surveys, an intensely cultivated agricultural zone. As a result, several critical areas were not investigated, among those are the regions from the ancient city of Umma (Tell Jokha) towards the Tigris tributary Shatt al-Gharraf and the area between the Tigris and the Shatt al-Gharraf, the plains between the ancient city of Girsu (Tello) and Lagash (Tell al-Hiba), as well as the region to the south-east to Nina (Tell Zurghul) and the area between Lagash (Tell al-Hiba) and Ur (Tell al-Muqayyar). In addition, the desert south and west of Eridu (Abu Shahrein), as well as the southern marshes, east of Ur and Lagash remained largely unknown archaeologically. Recent resurveying of parts of the region (Marchetti et al. 2019) as well as remote sensing surveys and the interpretation of declassified satellite imagery (CORONA etc.) confirmed that many sites had been missed (Hritz 2005, 2010). However, assessment based

[1] Abdulameer and I worked on the publication of this manuscript in the last month of his life when he was treated at Stony Brook Hospital in 2021. We were able to finish a very rough draft of the article before his passing on April 29th, 2022. When I returned to the manuscript nearly a year later, I had to make certain executive decisions with regards to the presentation of his survey data which I hope are in line with his wishes.

on field work remained very limited (Gasche and Tanret 1998; Marchetti et al. 2019; Wilkinson and Tucker 1995).

As the inspector (2003–2010) representing the State Board of Antiquities and Heritage of Iraq (SBAH) in Thi Qar province (Iraq), which contains most of the great Sumerian cities, including Uruk (Warka), Ur, Eridu, Larsa (Tell es-Senkereh), Lagash, Girsu, Nina, and Umma, I oversaw the protection of archaeological sites (Al-Dafar Al-Hamdani 2019, 2008a). Many sites in the regions have been and are still threatened by severe looting. The plan to develop a comprehensive and effective protection program, which ultimately entailed increasing the number of guards and installing an armed patrol (Al-Dafar Al-Hamdani 2008a) prompted me to undertake the survey of the entire Thi Qar province as well as parts of its neighboring provinces. The survey was carried out in stages between 2003–2010 and included surveying previously surveyed areas as well as the areas which had not been previously investigated. I was able to add some 1200 new sites to the combined inventory of sites surveyed by Adams (1981, 1972, 1965), Adams and Nissen (1972), Jacobsen (1960, 1969), Roux (1955) and Wright (1981), as well as those already presently mapped by the Department of Antiquities and Heritage of Iraq (SBAH) and published in the Atlas of Archaeological Sites in Iraq (1976).

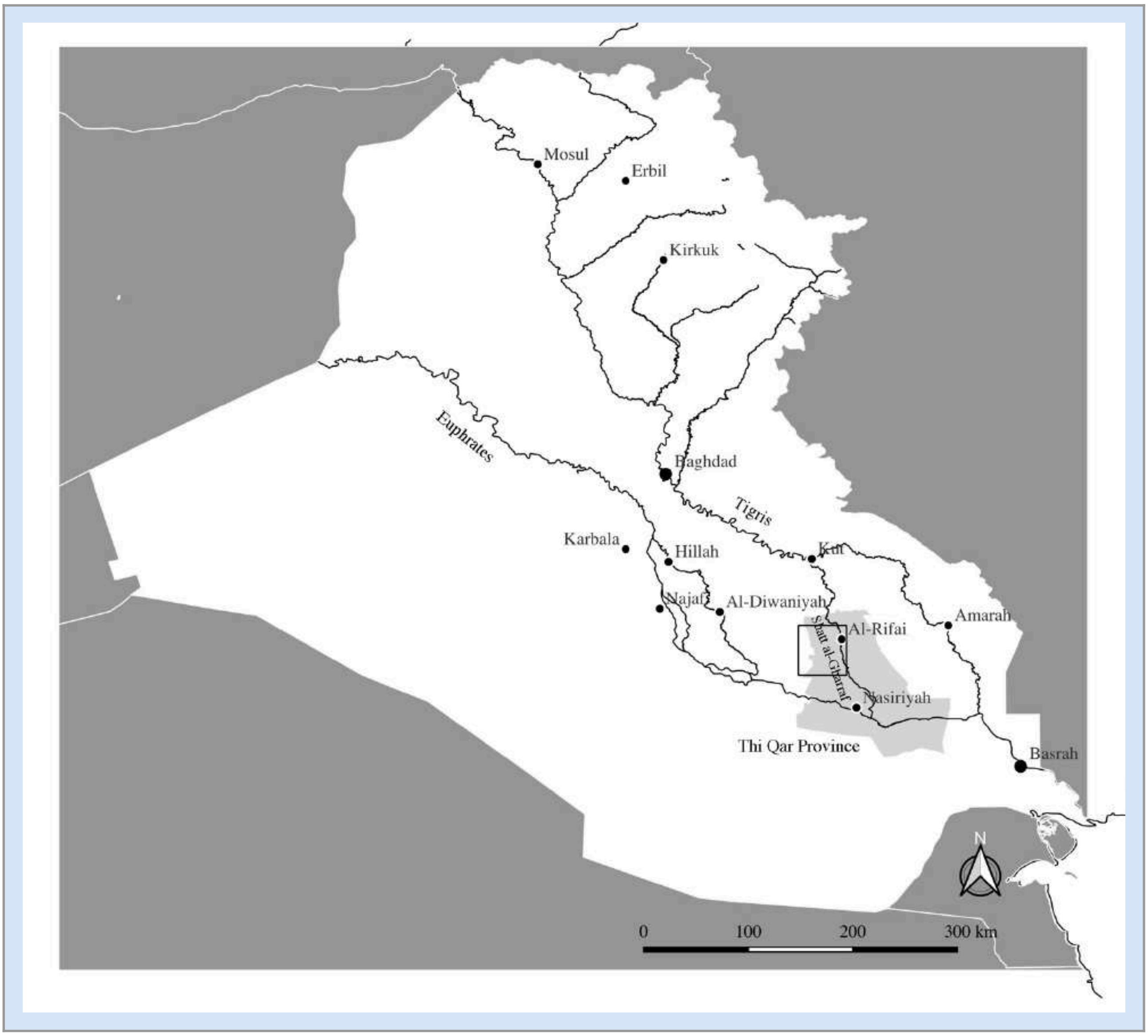

Figure 1: Location of the Umma region (black box)

This paper will focus on the Umma region located between Jidr in the north and Badtibira (Tell al-Medain/Medina) in the south and the general drainage area of the Shatt al-Gharraf to the east up to Tell Muhaliqiyat near the modern city Al-Rifai (see figure 1). Historically this region was of great importance, with the ancient city of Umma (Tell Jokha) as its most important site. The region was once home to one of the first city-states in the 3rd millennium BC and it shortly thereafter became an important province of the Akkadian kingdom (2350–2130 BC) (Frayne 1993, 261–262; Foster 1982). With the rise of the kingdom of the Third Dynasty of Ur, the Umma region was integrated as one of its major provinces (Dahl 2007; Sallaberger and Westenholz 1999; Steinkeller 2007, 1987).

This region is currently being resurveyed by Stephanie Rost and her team in the context of the Umma Survey Project (Rost and Di Michele 2022). The objectives of the Thi Qar Province Survey and the Umma Survey Project differ in such as the earlier was primarily focused on the protection of known and unknown archaeological sites. As looting had drastically increased during the Iraq War (2002–2011), surveying a large area as quickly as possible was of the utmost importance to halt the ongoing destruction of archaeological sites (Al-Dafar Al-Hamdani 2008b). Thus, time consuming intra-site analyses of individual sites was for the most part not attempted, which is the approach adopted in the Umma Survey Project to provide a detailed settlement history of the region. As the Umma Survey Project relies and builds on the results of the Thi Qar Province Survey, I wanted to make the results of the Umma region available to the public, particularly as the data collected provides a comprehensive map of confirmed archaeological sites in the region.

The survey efforts of Robert McCormick Adams (1981) and Hans-Jörg Nissen (Adams and Nissen 1972) covered only the western part of the Umma region, documenting 203 archaeological sites. The eastern part of the Umma region was at the time under active cultivation and hence inaccessible. The area has since gone out of agricultural use due to the lack of irrigation water and I was able to add close to 160 unknown sites to the inventory of known archaeological sites in the region (2008a, 2011, 2019). The higher efficacy of my survey in detecting sites can be attributed to several factors, ranging from certain areas having become more accessible to the adopting a different approach. This difference I feel warrants a more detailed discussion as it is relevant for future archaeologists working in the region and to highlight the urgency of more scientific collaborative efforts between foreign led projects and Iraqi archaeologists.

Foreign led projects, thus far, have relied on established survey methods that were geared towards the extensive investigation of large areas to detect and document as many sites as possible (Adams 1981, Adams and Nissen 1972). With the exception of the intensive surveys carried out at Kish (Tell al-Uhaimar), Mashakan-shapir (Tell Abu Duwari), Lagash, Ur and Uruk (Carter 1989; Finkbeiner 1991; Gibson 1972; Hammer 2022, 2019; E. C. Stone and Zimansky 2004), the surface record of individuals sites was often sampled in an opportunistic, unsystematic, and random fashion to establish basic parameters of a region's settlement history (Ur 2013, 133–134). Remote sensing has become an essential part of surveying efforts in Mesopotamia and has already been integrated early on by Adams (1981, 28–33) who made use of KLM aerial photographs as well as LANDSAT satellite imagery to aid the basic mapping of his survey areas. With the availability of high-resolution satellite imagery (e.g., CORONA, DigitalGlobe Quick Bird, HEXAGON, SPOT, SRTM) remote sensing techniques (alongside geographic information and positioning systems) have become one of the most important tools used in archaeological surveys (Hammer, FitzPatrick, and Ur 2022; Hritz and Wilkinson 2006; Hritz 2010; Ur 2016, 2003). It provides the much-needed bird's-eye perspective for an initial and remote reconnaissance of the study area in the development of a robust research designs and the planning of the field work stage. I have integrated it into my approach starting in 2008 to great effect (Al-Dafar Al-Hamdani 2015a).

However, the high resolution of satellite imagery can also lead to an overreliance on remote sensing techniques in the detection of archaeological sites at the expense of pursuing other sources of information. While remote sensing has greatly enhanced the chances of detecting and locating archaeological sites, the visibility of sites on high resolution imagery varies greatly depending on the year, season, and time of day the image was taken. To achieve total coverage of a survey area (Banning 2002, 167–168), which tends to be the goal in Mesopotamian archaeology to be able to locate places mentioned in the historical records, foreign and local archaeologists should be aware of where and from whom information about the location of ancient sites can be obtained. The success of locating ancient sites does greatly dependent on the level of familiarity with the local terrain as well as being able to tap into various local information networks, which are more accessible to Iraqi archaeologists from the region. In the following I want to outline the different venues that I pursued in locating and documenting in ancient sites that could function as a guide for future survey work in the region.

2. Detecting and documenting unknown sites

As I have previously outlined (Al-Dafar Al-Hamdani 2019, 2011, 2008a), the initiative to survey the entire Thi Qar Province was born out of my collaboration with the Carabinieri Command for the Protection of Cultural Heritage (Comando Carabinieri Tutela Patrimonio Culturale) in charge of safeguarding Iraq's archaeological and cultural heritage (Foradori 2017, 7–9; Russell 2008, 34–41). Thi Qar Province, containing many of the most famous ancient Sumerian cities, was particularly hard hit by looters, going back to 1994 (Russell 2008: 30–31). In 2003 the State Board of Antiquity and Heritage of Iraq (SBAH) began excavations at the ancient city of Umma (Tell Jokha) to protect the site, which had been one of the most central targets of looting in the region. This is when I began surveying Thi Qar province starting with the area directly surrounding Umma. I divided Thi Qar province into nine sections, each covering a specific region, which was systematically surveyed over the course of six years (see table 1).

Table 1

Subdivision of Thi Qar Survey Region

Section Nr.	Region
1	Umma and surroundings
2	Sites between Badtiberia and Larsa
3	Sites around Ur
4	Eridu and beyond to the west
5	Tell Lehem to the east – including the Hammar marshes
6	Central marshes
7	Al-Rasafa *aka* Rasafat Wasit and surroundings
8	Area around Girsu and Lagash
9	Area around Nina – eastwards and southwards.

Section 1–6 were located on the western side of the Shatt al-Gharraf and 7–9 was on the eastern side of the Shatt al-Gharraf. The basis for the division was a regional cadastre map (see below). We first surveyed Umma and an area from Tell Shmiet-Umma (Tell Jokha) -Umm al-Aqarib eastward towards Shatt al-Gharraf. In 2004 we moved slightly to the east to survey the area east of Shatt al-Gharraf from the modern towns of al-Rifai in the north and al-Shaṭra in the south and al-Diwaniyya in the east.

My method of locating sites was multifold. Prior to integrating remote sensing into my approach, the sources of information at my disposal were the Guidebook and Atlas of Archaeological Sites in Iraq

(Directorate General of Antiquities of Iraq 1976). The Guidebook contains information, such as the type of site (e.g., tell site, visible ruins of temples, palaces, castles, or fords), name of nearby village/s, and the date and issue of Iraq's national newspaper Al-Waqaei when a place was declared to be an archaeological site. Furthermore, the guidebook provides information about the chronology of the occupation of the site. The Atlas of Archaeological Sites of Iraq provides maps of the location of the various sites listed in the catalogue. However, those are of limited utility as the maps provide only very approximate locations. One issue, already noted by Adams (1981, 43), was that the files, which document visits and investigations of individual sites, kept by the Department of Antiquities (and which formed the basis of the Guidebook and Atlas) were archived based on a site's name rather than its location. The locations of sites on the other hand were transcribed from existing maps with variable accuracy and not independently confirmed. Adams (1981, 43) noted that the location of some sites that he revisited were at times off by 10 km or more.

A more reliable source to navigate the terrain and to and from archaeological sites were Iraqi military maps from 1986 of a scale of 1:1000000. Those maps had been updated based on satellite imagery in 2003 by the Iraqi army right before the Iraq War (2003–2011). I used those maps in combination with a cadastre map, dating to the 1960s of higher resolution (1:10,000 m) which document next to the location, size, and ownership of agricultural land, also the location of archaeological sites within those land holdings. The initial cadastral survey was carried out under the British occupation enacting the "Law of Settlement of Land Rights" in 1932, with the goal to register land and settle land disputes (Hashimi and Edwards 1961, 72–73). Another land registration or modification thereof followed the "Agrarian Reform Law" which was enacted in October of 1958 by the new government following the 1958 revolution led by Abdul Karim Qassim. The law imposed a ceiling of the size of privately owned land, redistributed the seized land to landless farmers, created agricultural cooperatives and defined minimum wages for agricultural workers. In addition, it regulated tenancy relationships and imposed contracts between landlord and tenant set to three years following renewals (Abid 1995, 22–23; Baali 1966, 58–61; Farouk-Sluglett and Sluglett 2003, 76; Hashimi and Edwards 1961, 75–78). No land reform has been carried out since and the size and extent of individual cadastres have not changed.

In 1990s a commission of the State Board of Antiquities and Heritage of Iraq (SBAH) approached the Ministry of Agriculture to include an archaeological assessment of agricultural land, and of any development projects (e.g., houses, commercial complexes or public buildings, roads, irrigation canals, pipelines, etc.) to detect, record and protect sites from destruction (see LAW No.55 of 2002 For the Antiquities & Heritage of Iraq published by UNESCO). From this time onward, the issuing of building permits or approval of changes in the management of agricultural land would trigger an archaeological investigation. In the case of agricultural land, a memorandum stating the presence and size of archaeological sites was added to the updated land titles to protect archaeologically sensitive areas. Furthermore, the location of archaeological sites was recorded on existing cadastre maps. Farmers are prohibited by law to cultivate archaeological sites and regular investigations by officials of SBAH would ensure that those regulations were followed.

Even though these cadastre maps formed an important source of information for my survey, it became quickly apparent that the number of sites in certain areas was grossly underreported. As for example in one cadastre (Nr. 23) to the east of Al-Rifai, only two archaeological sites were recorded, however upon my visit I registered a total of 24 sites of various sizes. Local farmers with their intimate familiarity of the surrounding landscape, are an important source of information to locate archaeological sites, particularly those that are not visible on satellite imagery. Thus, any archaeologist should try to set time apart to build relationships with the local populations they encounter in the areas they work. Another important source of information were conversations with local civilian guards, who were recruited from nearby villages and employed by SBAH starting in the 1970s to monitor and protect archaeological sites. Monthly stipends were paid in exchange for monthly reports on the state of the archaeological sites under their watch. The stipends from SBAH became also a much sought-after source of income among the farming population which had suffered extreme hardship following the economic sanctions imposed by the United Nations Security Council in 1990 following the invasion of Kuwait. The sanctions had a particularly adverse effect on Iraq's agricultural sector and production was severely reduced, particularly in the south of Iraq. Dwindling water supply resulting from an ill-maintained and ill-managed supply system robbed many farming communities of their income (Alnasrawi 2001, 208–214). Hence farmers would approach representatives of SBAH reporting previously undocumented archaeological sites in the hope of being installed as permanent civilian guards. If the site/s warranted protection their wish was often granted. Moreover, farmers seeking employment was (and still is) an important source of information to locate and register unknown archaeological sites. By 2003, about 112 civilian guards were installed all over Thi Qar Province. In my experience, the familiarity with the local terrain and landscape of these local guards is pivotal in locating as well as for navigating to and from archaeological sites. Other sources of information were social events and encountering sites when travelling around as well as social media and informal exchanges in villages and markets.

3. Surveying methods

As mentioned earlier my survey efforts were driven primarily by cultural heritage preservation and between 2003-2005 my survey was focused on documenting sites, detecting looting, and implement a protection plan. From 2005 onwards, the objective of my survey included also determining and describing the nature of ancient occupation of Thi Qar province and its settlement history. I visited the site either alone or with a team of two of three colleagues. Depending on the size of the team, the site was divided into equal parts to be surveyed by individual archaeologists. We sampled the surface record for diagnostic pottery pieces and artifacts (e.g., cylinder seals, figurines, stamped bricks, sickles, stone lithics, fragments of bronze or metal tools, or coins). The material collected from the sites was stored in the Nasiriyah Museum for future analysis. However, in the chaos of the Iraq War (2002–2011) war, the collection got lost. Thus, I was not able to confirm my initial dating of the sites by means of a close study of the collected surface material. While I am confident about the accuracy of the general trends in the settlement history that I was able to describe based on my observations, I fully expect that future re-surveying of the area will include a correction in the details of the chronology of the occupation of individual sites. However, each site on the maps presented here is a confirmed archaeological sites and hence a very reliable guide for future investigations.

The initial documentation of archaeological sites was done in an Excel sheet, recording the coordinates measured with a handheld Toshiba GPS tracker (provided by the Italian Carabinieri), the modern name of the site, the chronology of the occupation of the site, names of nearby village, information on the state of preservation and the presence or absence of looting, names of civilian guards if applicable, and a general description of the site. Since 2008, I included remote sensing into my approach. This included an initial remote sensing inspection of specific areas using DigitalGlobe QuickBird imagery and mapping archaeological sites and ancient landscape features, that I then visited on the ground. In the remote sensing analysis, I observed certain patterns that are of importance for further detection of unknown archaeological sites. For one, looting holes are very visible on satellite imagery and provide unquestionable proof of the existence of a sites (see also Marchetti et al. 2019, 215). Furthermore, archaeological sites, particularly those with a visible elevation are preferred locations for the construction of religious shrines, cemeteries, electricity towers or even schools as they are beyond the reach of floodwater. Sites tend not to get settled, except for those located in the marshes as they provide solid and dry ground within a wetland. Outside of that, building domestic structures on top of archaeological sites is usually avoided as the soil tends to be relatively unstable and out of superstition that souls inhabiting ancient graves could unintentionally be disturbed and cause harm.

In my classification of site size and type, I relied on the typology defined by Adams and Nissen (1972, 18) which considers sites ranging between 01.– 6 ha as villages, sites 6.1–25 ha in size as towns and sites larger than 50 ha as urban centers. Adams (1972, 71–75) considers sites above 10 ha as urban. These site categories are, in my opinion, too coarse, lumping different settlement types into one category. For example, there is ample evidence of special purpose sites of less than 2 ha with a single public building with a very specific function (e.g., Tell Khaiber, Campbell et al. 2017). Adams's site size category does not account for them and hence does not fully describe the diversity of site types. Thus, my categorization distinguishes between seven site classes (see table 2): Sites of less than 2 ha large are either special purpose sites or hamlets. Sites ranging between 2–4 ha in size are small villages, while large villages range between 4–8 ha. I consider sites ranging between 8–16 ha small towns and distinguish them from regular towns whose size ranges between 17–25 ha. I would consider sites ranging between 25–50 ha substantial towns but would consider sites larger than 50 ha as cities.

Table 2

Settlement classes

Size in ha	Type of settlement	Size category
> 2	Special purpose site/ hamlet	1
2–4	Small village	2
5–8	Large village	3
9–16	Small town	4
17–25	Town	5
26–50	Large town	6
< 50	City	7

4. Results of the Survey

I recorded a total of 179 sites for the Umma region, of which about 15% (27 sites) have been already documented by Adams (1981) and Adams and Nissen (1972). These include the most prominent sites, such as Farwa, Tell Bizikh (Zabalam), Tell al-Medain (Badtibira), Tell Jokha (Umma), Umm Aqarib, Umm Ajaj, and Tell Shmiet. Resurveying previously surveyed areas (Adams 1981; Adams and Nissen 1972) showed that many small and even medium sites had been missed, as the motorized survey method tends to favor the discovery of large sites (see also Ur 2013, 133–134; Marchetti et al. 2019, 215). Adams (1981, 46–47) was very cognizant about the biases and limitations of his approach and always emphasized the provisional nature of his results. He conducted a formal assessment to quantify the effectiveness of his approach by resurveying a 100 km^2 in the vicinity of Nippur (Nuffar) based on a 1 km grid. Based on the results, he concluded that the number of sites he missed was at least in the range of one-third, particularly smaller sites with very modest elevations (Adams 1981, 40–42). Thus, he always emphasized the necessity of re-surveying those areas more intensively and systematically to correct the shortcomings of his initial surveys.

Table 3

Number of settlements in different site size categories

	Hamlet (> 2 ha)	Sm Village (2–4 ha)	Lg Village (4–8 ha)	Sm Town (9–16 ha)	Town (17–25 ha)	Lg Towns (26-50 ha)	Cities (<50 ha)	Total
Ubaid 5000–4000 BC	3	1	6	2	0	0	0	12
Uruk 4000–3100 BC	2	5	6	4	1	0	0	18
Jemdet Nars 3100–2900 BC	2	10	8	6	1	1	1	29
Early Dynastic I 2900–2600 BC	17	20	19	6	1	1	2	66
Early Dynastic II/III 2600–2350 BC	21	26	16	11	3	2	6	85
Akkadian 2350–2120 BC	19	19	13	8	5	2	5	71
Ur III/Isin-Larsa 2112–1850 BC	30	33	20	11	5	2	7	108
Old Babylonian 1850–1595 BC	12	21	11	7	3	2	6	62
Kassite 1475–1155 BC	1	6	4	1	1	0	1	14
Middle Babylonian 1158–1026 BC	0	0	0	0	0	0	0	0
Late Babylonia 1026–911 BC	0	0	0	0	0	0	0	0
Assyrian 911–619 BC	0	0	0	0	0	0	0	0
Neo-Babylonian 620–539 BC	12	14	12	6	3	2	1	50
Achaemenid 539–330 BC	5	6	3	3	2	1	1	21
Seleucid/Parthian 312 BC–AD 224	12	9	8	4	3	1	4	41
Sasanian AD 224–651	18	19	9	7	3	2	4	61
Early Islamic AD 638–1000	17	13	7	4	2	2	1	46
Middle Islamic AD 1000–1517	21	15	10	3	1	1	1	52
Late Islamic AD 1517–1924	20	17	11	4	1	1	2	56

Despite this, the data from the Thi Qar Survey in the Umma Region confirmed most of the general trends in the settlement history that have been documented by Adams (1981) and Adams and Nissen (1972) farther to the west (see table 3). For one, the Umma region also experienced a remarkable urbanization during the Early Dynastic II/III period (2600–2350 BC) with a vast majority of people living in small urban centers and cities as has been documented in other regions of southern Mesopotamia (Adams 1981, 138–141; Adams and Nissen 1972, 11–12, 17) as well as in northern Mesopotamia (Ur 2010, 404–412). Furthermore, the two apogees of the greatest extent of the settled area (see figure 2) also fall into the Ur III/Isin-Larsa period (2112–1850 BC) and the Sasanian period (AD 224–651). As was the case in the Nippur-Uruk region (Adams and Nissen 1972, 39–41; Adams 1981, 165–170), the Umma region was largely abandoned during the late second to the mid-first millennium BC due to shifting watercourses that robbed the region of water. I was able also to confirm a resettlement of parts of the Umma region in the first millennium BC and AD that went along with the artificial manipulation of the rivers to restore water supply to the region (see also Adams 1981, 175–179). The settlement data of the Umma region also indicated a drastic urban decline during the Early to Late Islamic period (AD 638–1924) and increased ruralisation of the region that has also been documented by Adams (1981, 183–185) in the central alluvial plain.

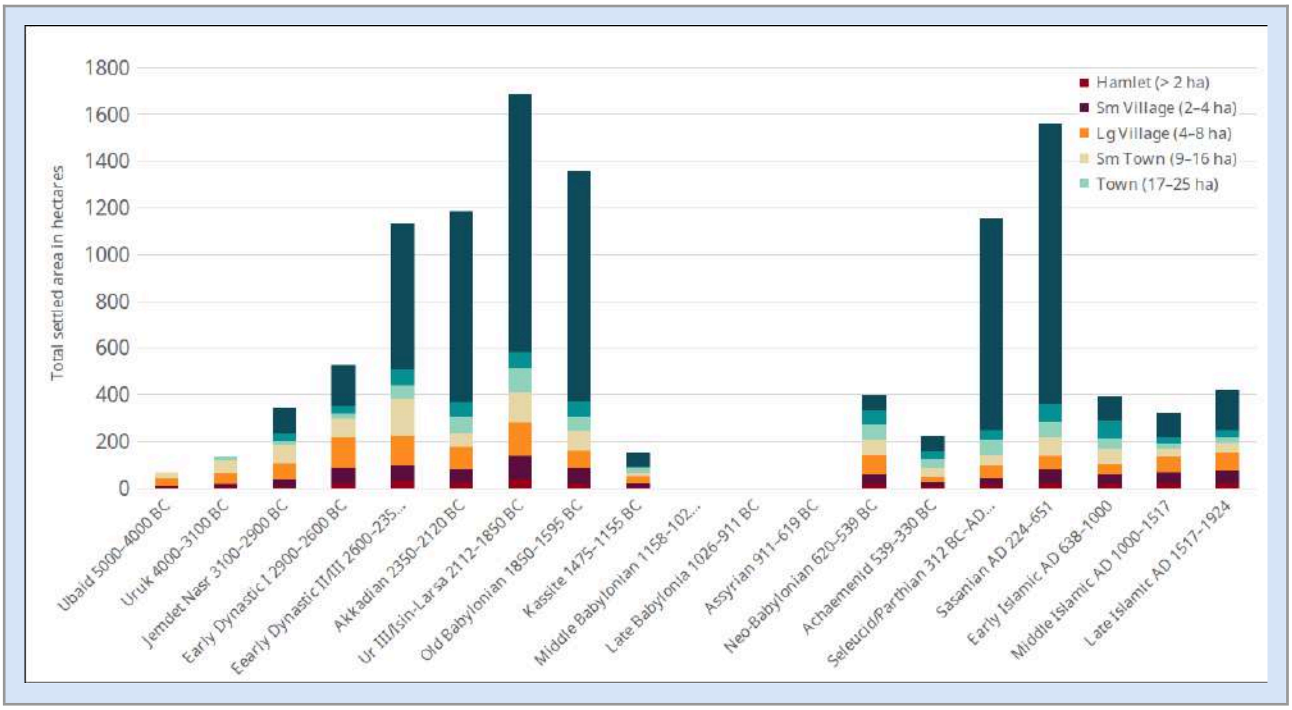

Figure 2: Number of settlements in different site size categories

Despite those similarities, there are also considerable differences in the trends observed in the Thi Qar Survey data for the Umma region when compared to those collected by Adams (1981) and Adams and Nissen (1972) in the Nippur-Uruk Region. For one, there is a much less drastic increase in the number of sites (see table 3 and figure 2) and the total occupied area (see table 4 and figure 3) between the Ubaid (5000–4000 BC) and the Uruk (4000–3100 BC) period when compared to the Nippur-Uruk region, which was most pronounced in the number of small village of less than 4.5 ha in size (Adams and Nissen 1972, 18, Adams 1981, 69–70; see also Ur 2013, 149, Fig. 7.9). A drastic increase in the number of sites occurred in the Umma region much later between the Jemdet Nasr (3100–2900 BC) and the following Early Dynastic I period (2900–2600 BC). The expansion of total settled area, however, was modest as the increase in number occurred primarily in smaller sites, such as hamlets (> 2 ha), small (2–4 ha) and large villages (5–8 ha) (see figure 2, 3 and table 3, 4). This trend got completely reversed in the following Early Dynastic II/III period (2600–2350 BC), where a 24% increase in the total number of sites occurred, but the occupied area nearly tripled (see figures 2, 3 and table 3 and 4). This increase is due to several major cities such as Umma, Umm Aqarib, Zabalam, Tell Shmiet and Tell Jidr rising to political importance. Most of those cities had been settled much earlier, however the extend of occupation in early periods is often not known and/or poorly attested (Adams and Nissen 1972, 219–220, 226–228). As was recorded farther to the west, the vast majority (75%) of the population of the Umma region now lived in small urban centers (<17 ha) and major cities (<50 ha).

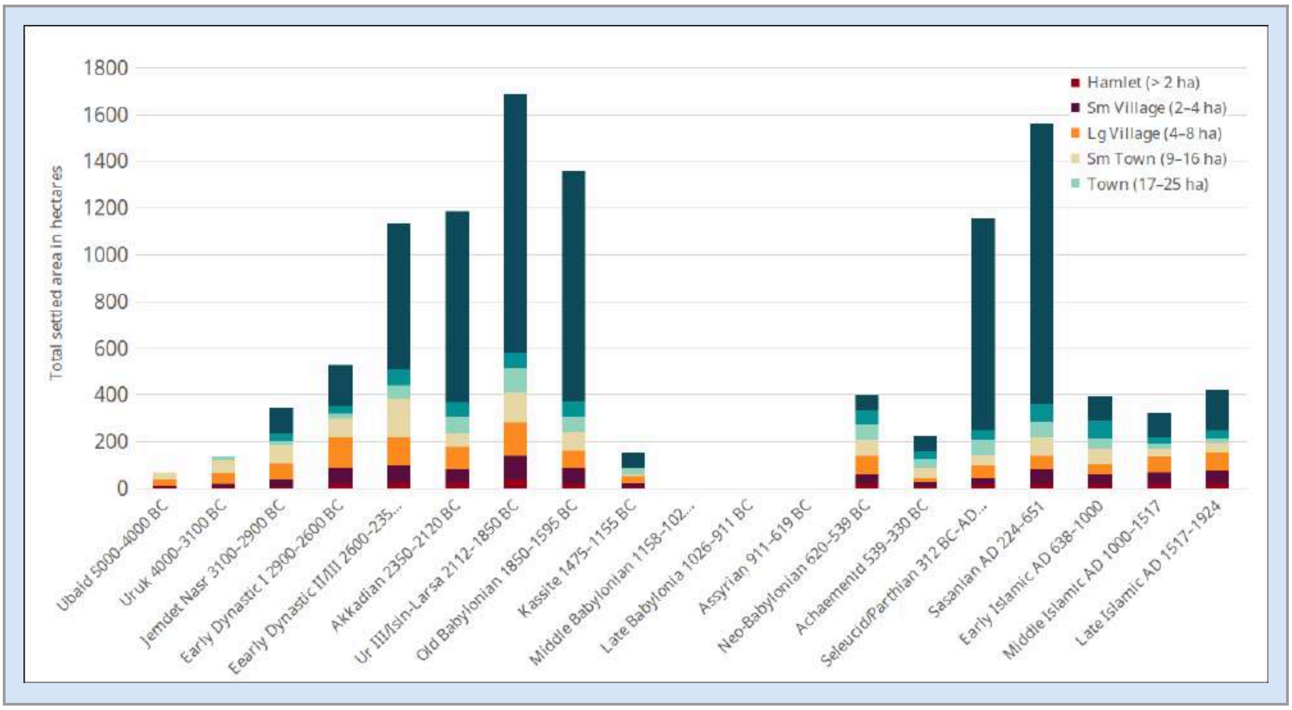

Figure 3: Total occupied area of different site size categories

The growth of cities during the Early Dynastic II/III period (2600–2350 BC) in other parts of southern Mesopotamia occurred, according to Adams (1981, 138–141; see also Adams and Nissen 1972, 11–12) “at the expense of smaller rural settlement in their hinterland” (Adams and Nissen 1972, 17). This trend has also been confirmed in the resurvey of the area just east and north of the Umma region by the QADIS survey carried out in 2016–2019 under the direction of Nicolò Marchetti and his team (Marchetti et al. 2019). The data from the Umma region do not confirm that pattern. While the increase in sites was most noticeable in the urban category (small, regular, and large towns and cities), the number of rural settlements (hamlets and small villages) also grew, except for large villages (5–8 ha) whose numbers slightly dropped (see figure 2 and table 3). As the rural depopulation is not attested in the Umma region, its urban growth must have relied on an influx of people from elsewhere. This result, however, also raises the question whether the rural depopulation observed by Adams (1981) and Adams and Nissen (1972) may in reality be the result of their approach, which was less sensitive to detecting smaller sites. Future resurveying of those areas will answer that question.

The greatest extent of settled area in the Umma region falls into the Ur III/Isin Larsa period (2112–1850 BC) and not, as in the Nippur-Uruk region, in the Sasanian period (AD 224–651, Adams 1981, 180–183). This might be related to the fact that the Umma region, being at the margin of the Sasanian Empire was far less urbanized compared to the Ur III/Isin-Larsa period. Adams (1981, 182) also noted this discrepancy between southern Mesopotamia and the Diyala region with the latter having much larger population aggregates due to its proximity to Ctesiphon, one of the major Sasanian urban centers in Mesopotamia. The Umma region, being far removed from any Sasanian political centers, was predominantly rural, with 74% of the settlements falling into the hamlet, small and large village categories (see table 4). In the following I want to describe the details of datasets for each period:

Table 4

Percentage of total occupied areas of different settlement types

	Hamlet (> 2 ha)	Sm Village (2–4 ha)	Lg Village (4–8 ha)	Sm Town (9–16 ha)	Town (17–25 ha)	Lg Towns (26–50 ha)	Cities (<50 ha)	Total area in ha
Ubaid 5000–4000 BC	5.16	5.19	47.78	41.86	0	0	0	66.361
Uruk 4000–3100 BC	1.38	10.96	33.26	41.33	13.08	0	0	131.691
Jemdet Nasr 3100–2900 BC	0.75	9.64	19.14	23.67	5.02	9.53	32.25	343.27
Early Dynastic I 2900–2600 BC	3.92	11.34	25.87	15.02	4.19	6.2	33.45	527.038
Early Dynastic II/III 2600–2350 BC	1.83	4.8	8	10.21	4.19	4.94	65.96	1403.87
Akkadian 2350–2120 BC	1.86	4.57	7.62	7.51	7.78	5.25	65.42	1246.243
Ur III/Isin-Larsa 2112–1850 BC	2.21	6	8.4	7.67	6.18	3.88	65.67	1684.65
Old Babylonian 1850–1595 BC	1.15	4.98	5.59	5.98	4.9	4.83	72.57	1353.186
Kassite 1475–1155 BC	1.07	12.98	18.36	7.14	16.69	0	43.76	149.875
Middle Babylonia 1158–1026 BC	0	0	0	0	0	0	0	0
Late Babylonia 1026–911 BC	0	0	0	0	0	0	0	0
Assyrian 911–619 BC	0	0	0	0	0	0	0	0
Neo-Babylonian 620–539 BC	3.37	11.25	20.44	16.44	17.08	14.84	16.58	395.517
Achaemenid 539–330 BC	2.4	8.56	8.43	17.52	19.43	13.7	29.96	218.912
Seleucid/Parthian 312 BC–AD 224	1.04	2.41	4.66	4.27	5.13	3.89	78.6	1155.05
Sasanian AD 224–651	1.13	3.91	3.64	5.27	4.13	4.8	77.11	1559.893
Early Islamic AD 638–1000	4.2	10.56	10.95	16.92	11.27	19.1	27	392.176
Middle Islamic AD 1000–1517	6.35	14.26	20.52	9.8	6.93	9.3	32.84	322.495
Late Islamic AD 1517–1924	4.76	12.93	17.9	10.56	5.37	7.22	41.25	415.672

5. Ubaid Period (5000-4000 BC)

I was able to record 12 settlements dating to the Ubaid period (5000–4000 BC), all of which remained settled in the following Uruk period (4000–3100 BC) and/or later (see figure 4 and table 3). The occupation of the Umma region during the Ubaid period (5000–4000 BC) was sparse, which corresponds to the settlement data of the Nippur-Uruk region (Adams and Nissen 1972, 9–11; Adams 1981, 59–60) and the Ur-Eridu basin (Wright 1981, 123–125). The settlement density in the Ur-Eridu basin and around Uruk was slightly higher than in the Nippur–Abu Salabikh region which, according to Adams (1981, 59–60), could be the result of greater alluviation that may have obscured small sites.

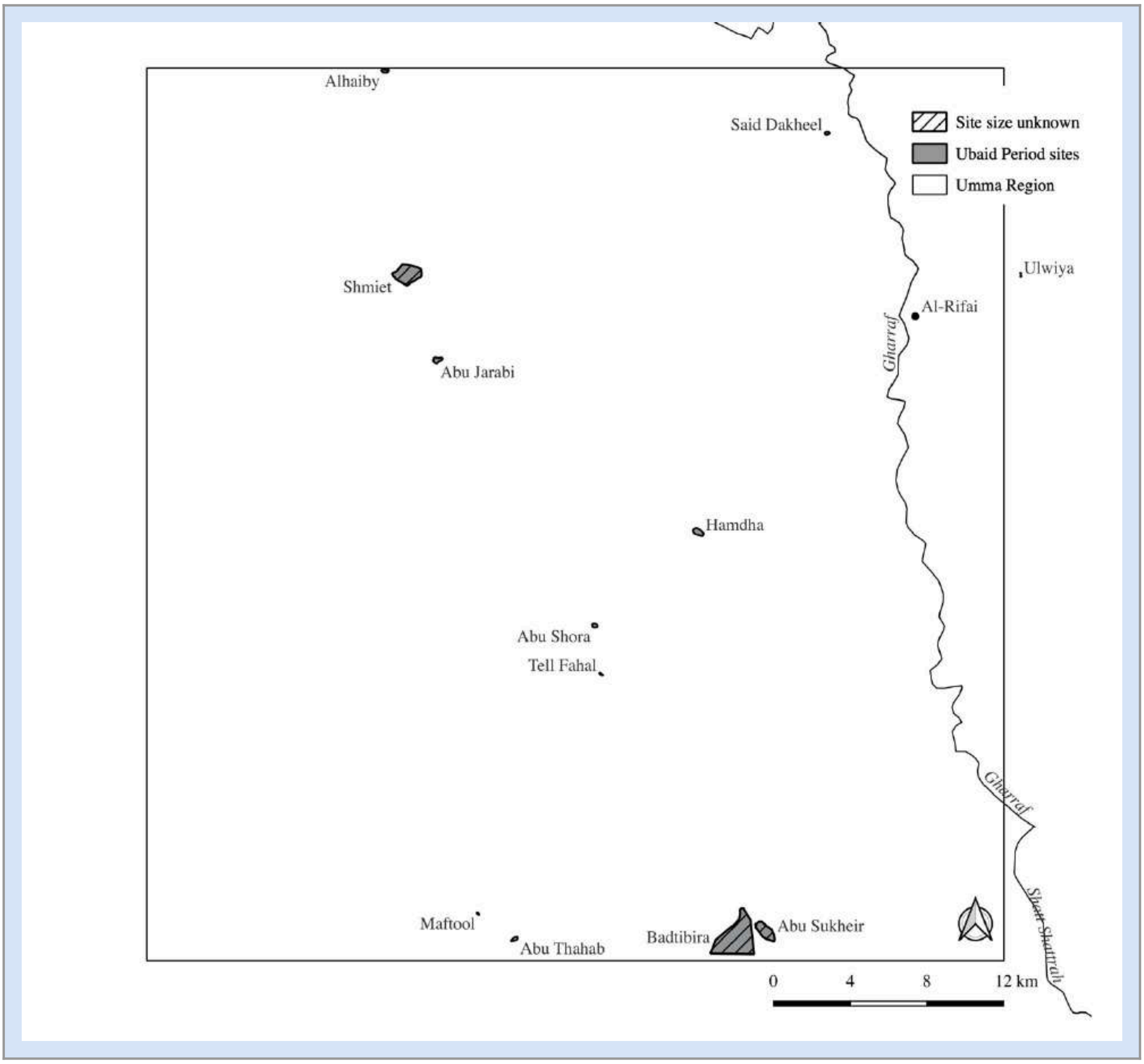

Figure 4: Ubaid period (5000–4000 BC) occupation of the Umma region

In addition, overburden from later occupation made it difficult at times to estimate the extent of the Ubaid occupation for most sites. In the Umma region most of the Ubaid sites were small, with four of them being hamlets and small villages, while six constitute large villages (4–8 ha). Two sites, Abu Jarabie (11 ha) and Hamdha (16 ha), are considerably large and fall into the category of small towns. The only other Ubaid sites in southern Mesopotamia of that size were Ur and Eridu, ranging between 10–12 ha (Wright 1981, 123–125). Abu Jarabie was covered by a sand-dune belt when I conducted the Thi Qar survey, which has since moved. The site was resurveyed by Rost and her team (Rost and Di Michele 2022). Hamdha at the time was located in a cultivated area that had since gone out of use and future resurveying might provide additional insights into the nature of Ubaid sites in the Umma region. The sites Hamdha, Abu Shora, Tell Fahal are located at the edge of a visible depression which contained freshwater shells on the surface, which may have been a marsh area in Ubaid times. In addition, both Abu Jarabie and Hamdha have a visible mount that can be seen from a distance, and which may indicate that they have been located on elevated ground within a wetland. Pournelle (2007) and Pournelle and Algaze (2014) have argued that the exploitation of maritime and marsh resources may explain the higher settlement density in the very south. Pournelle (2007, 34, 58) further argues that having access to maritime and marsh resources, next to agriculture, was instrumental in the development of socio-political complexity and the onset of urbanism. The fact that two

Ubaid sites in the Umma region could be considered small towns, both located on high ground, possibly in the midst marshland would support this argument. In addition, most Ubaid villages in the Umma region were large, further supporting the fact that the mixed economy of relying on agriculture and the exploitation of marsh resources sustained sizable population aggregations.

6. Uruk Period (4000–3100 BC)

The Umma region did not experience a sudden increase in the number of sites and population as did other regions in the central alluvial plain (Adams 1981, 60–61, Adams and Nissen 1972, 11; see also Marchetti et al. 2019, 222–223). I was able to record 18 sites dating to the Uruk period, of which nearly half had already been occupied in the preceding Ubaid period (5000–4000 BC). Newly founded Uruk sites were primarily small villages (2–4 ha), small (9–16 ha) and large (17–25 ha) towns (see table 3). The latter reflects the onset of a process of urbanization which was particularly notable at Uruk, which reached a size of 250 ha by the end of the Uruk period (Finkbeiner 1991, 191–195). The largest Uruk period site in the Umma region is Gubaiba, located just south of the modern city of Al-Rifai, and constituted, with a size 17 ha a town. As can be seen on the map (see figure 5) most of the recorded Uruk period sites fall into a narrow corridor running from northwest to southeast, which may indicate the presence of an ancient watercourse running from the site Tell Shmiet to Umma Aqarib. Noteworthy is a cluster of four sites (Hefaina [3.3 ha], Um Halfawiya [16.2 ha], Hulail [3.8 ha] and Thigal [2.5 ha]) that are all located within a 1.5 km radius. The clustering of sites with a two-tiered settlement hierarchy appears to be a hallmark, particularly in the Late Uruk period (3300–3100 BC), according to Adams and Nissen (1972, 25–28), and was particularly apparent in the vicinity of Uruk. Whether or not Um Halfawiya was a local center to the smaller subsidiary sites remains to be determined but the proximity of those sites suggests some interdependency. Even though my data do not confirm a pronounced increase in the total numbers of settlements and total occupied area, I was able to confirm a clear settlement hierarchy, that was recorded by Adams and Nissen for the Uruk-Nippur region (1972, 18), with every settlement category present from small hamlets (>2 ha) to small and regular sized towns (9–25 ha).

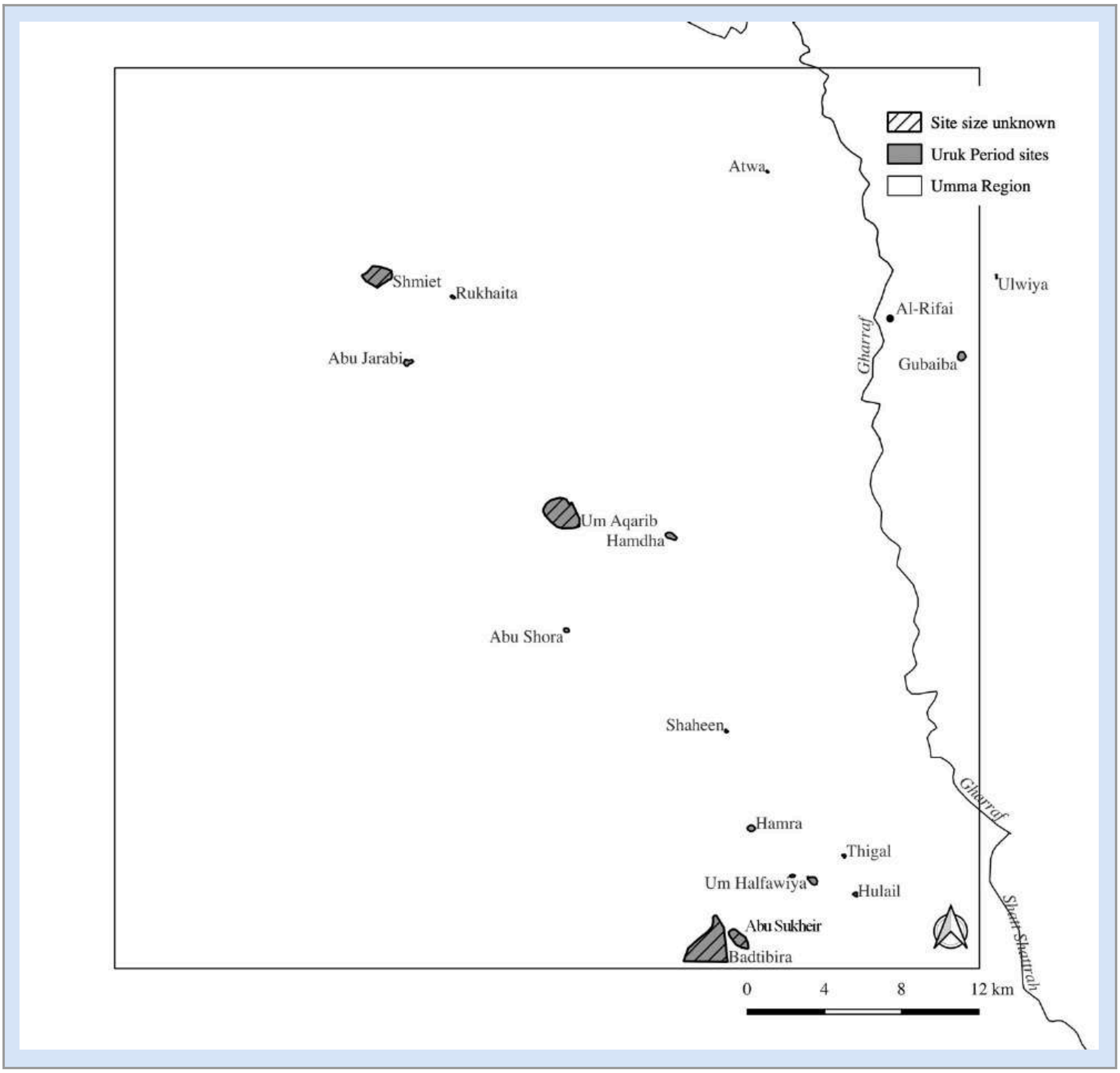

Figure 5: The Uruk Period (4000–3100 BC) occupation of the Umma region

7. Jemdet Nasr (3100–2900 BC)

The Umma region became fully urbanized during the Jemdet Nasr period with Um Ajaj (110.7 ha) reaching urban dimensions (see figure 6). There is a considerable continuation in occupation of the region between the preceding Uruk period (4000–3100 BC) into the Jemdet Nasr period. twelve out of the 29 Jemdet Nasr-Period sites were already settled in the Uruk period. Newly founded settlements were primarily small villages (2–4 ha), which doubled in number, while there was only a slight increase in the number of large villages and small towns (see table 3). The two urban sites Mansuriyah (32.6 ha) and Um Ajaj (110.7 ha) are of great interest as they appear to have been occupied for a relatively short period of time. Founded in the Jemdet Nasr period they remain occupied only in the preceding Early Dynastic I (2900–2600 BC) and II/III period (2600–2350 BC). Both sites have been visited by Adams and Nissen (1972, 228, 235) who confirmed the timeframe of occupation. While it is possible that these sites might have been much smaller at their initial foundation in the Jemdet Nasr period, it is certain that they grew substantially in the Early Dynastic period (2900–2350 BC). Um Ajaj was a major urban site in the Early Dynastic I period (2900–2600 BC). The other four sites (Abu Sukheir, Tell Shmiet, Um Aqarib and Badtibira [Tell al-Medain]), which are known urban sites of the subsequent periods, were already inhabited in the Jemdet Nasr period, however the extent of that occupation could not be determined.

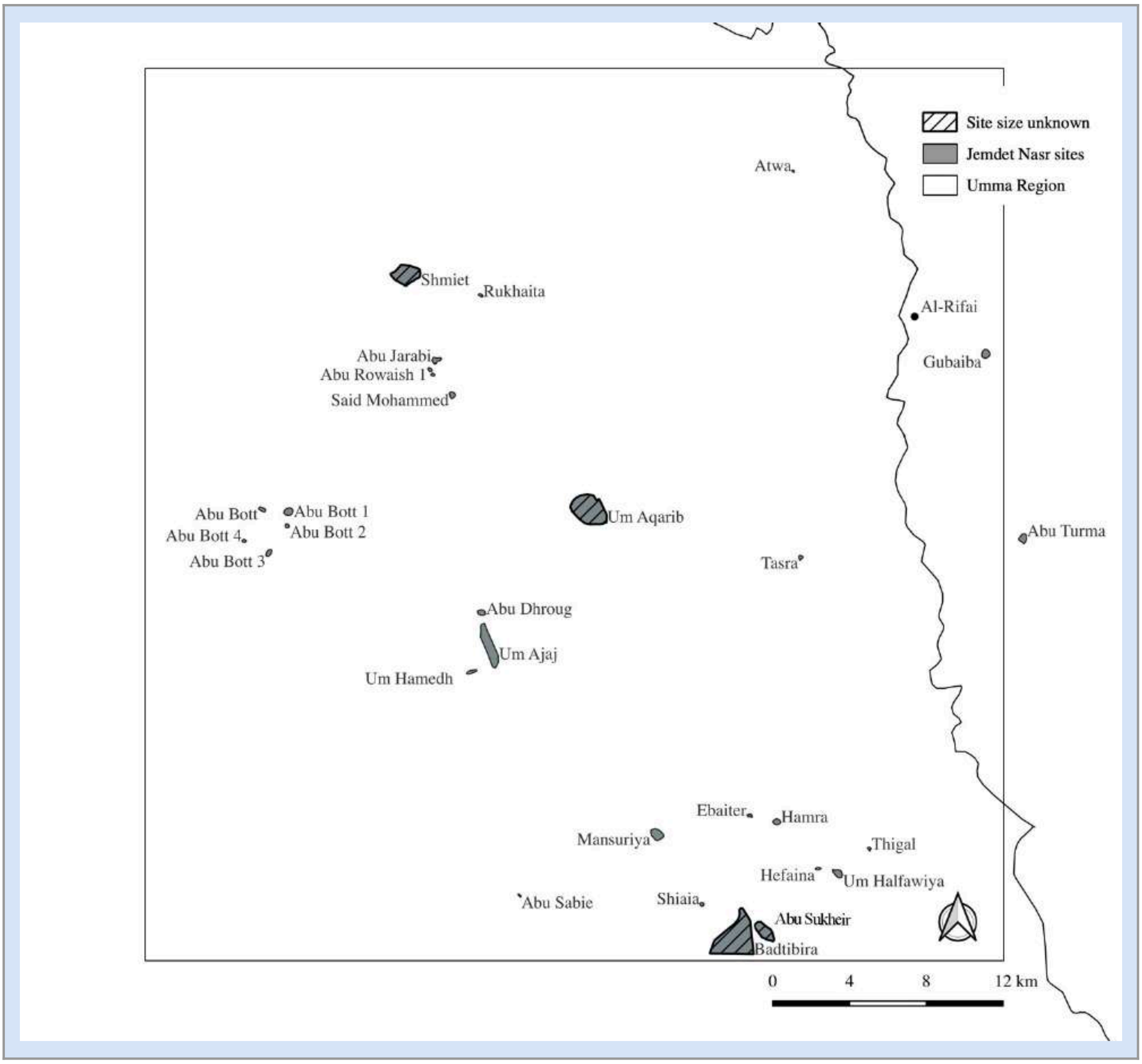

Figure 6: The Jemdet Nasr period (3100–2900 BC) occupation of the Umma region.

The settlement pattern is now less concentrated along the northwest-southeast corridor but fans out on either side to cover the Umma region more evenly (see figure 6). There are several noticeable clusters of sites, Abu Bott 1–4, Abu Jarabie, Abu Rowaish, and Said Mohammed and eight small and large sites surrounding Badtibira (Tell al-Medain). The Abu Bott cluster and the sites surrounding Badtibira have been visited by Adams and Nissen (1972, 227, 235) whose dating of the sites I could confirm. Abu Bott 1 appears to be located alongside an ancient watercourse running from north to south whose traces are still visible on current satellite (Bing Aerial) imagery. Given the occupation of the sites, this watercourse was most likely active between the Jemdet Nasr (3100–2900 BC) until the Old Babylonian period (1850–1595 BC).

It remains to be determined whether the Abu Bott cluster is one large site with smaller subsidiary sites. When combined, Abu Bott 1 (16 ha) and 2 (4 ha) would reach the size of a major town. Abu Bott 3, with 8.6 ha being the second largest site in the cluster was heavily looted in the 1990s. Most looting holes that I encountered in 2004 were visibly old (filled in and overgrown). The fact that this site was singled out by looters suggest that it may have housed public buildings with many marketable artifacts. Abu Bott 3 was located along the river, with a canal leading off to a small harbor, which suggests that the site could have functioned as a control point to manage river transport, collecting taxes and/or housing storage facilities, public and administrative buildings. Abu Bott 4 (2.8 ha) on the other hand was most likely a special-purpose site. The surface collection was heavily dominated by overfired slag and might have been a pottery production site. Moreover, Abu Bott 4 (also seen at Um Ajaj) has visible holloways around the site that have been documented and extensively studied in Upper Mesopotamia (Wilkinson 2003, 60; Ur 2002, 2003). As has been suggested by Ur (2003, 110–111) holloways ways are remnants of human traffic between settlements and agricultural fields and their surroundings as well as moving flocks of animals to pasture. Given that the landscape in southern Mesopotamia was interspersed with marshlands, the use and function of hollow ways may have been slightly different.

8. Early Dynastic I Period (2900–2600 BC)

All 29 sites that date to the Jemdet Nasr period (3100–2900 BC) continue to be inhabited in the following Early Dynastic I period. The Umma region experienced a drastic expansion of total occupied area (see figure 3 and table 4) and an increase in the total number of settlements (see figure 2 and table 3), which are now more evenly spread across the area (see figure 7). The total number of sites more than doubled (from 29 to 66) and new foundation of settlements are found in the smallest site size category of less than 2 ha. The number of hamlets and special purpose sites rose from two in the Jemdet Nasr period to 17 in Early Dynastic I period (see table 3 and figure 2). The number of small (2–4 ha) and large villages (5–8 ha) also doubles, while the number of small, regular, and large towns remained the same, though they may have grown in size from the preceding Jemdet Nasr period. The considerable increase in hamlets, small and large villages could indicate the expansion of farming, particular of irrigation agriculture. As has been observed by Adams (1981, 160–164) farther to the west of the Umma region, with the onset of the Early Dynastic period (2900–2350 BC) settlements tended to be more strictly aligned along watercourses, indicating the necessity to have access to water with the expansion of irrigated grain cultivation.

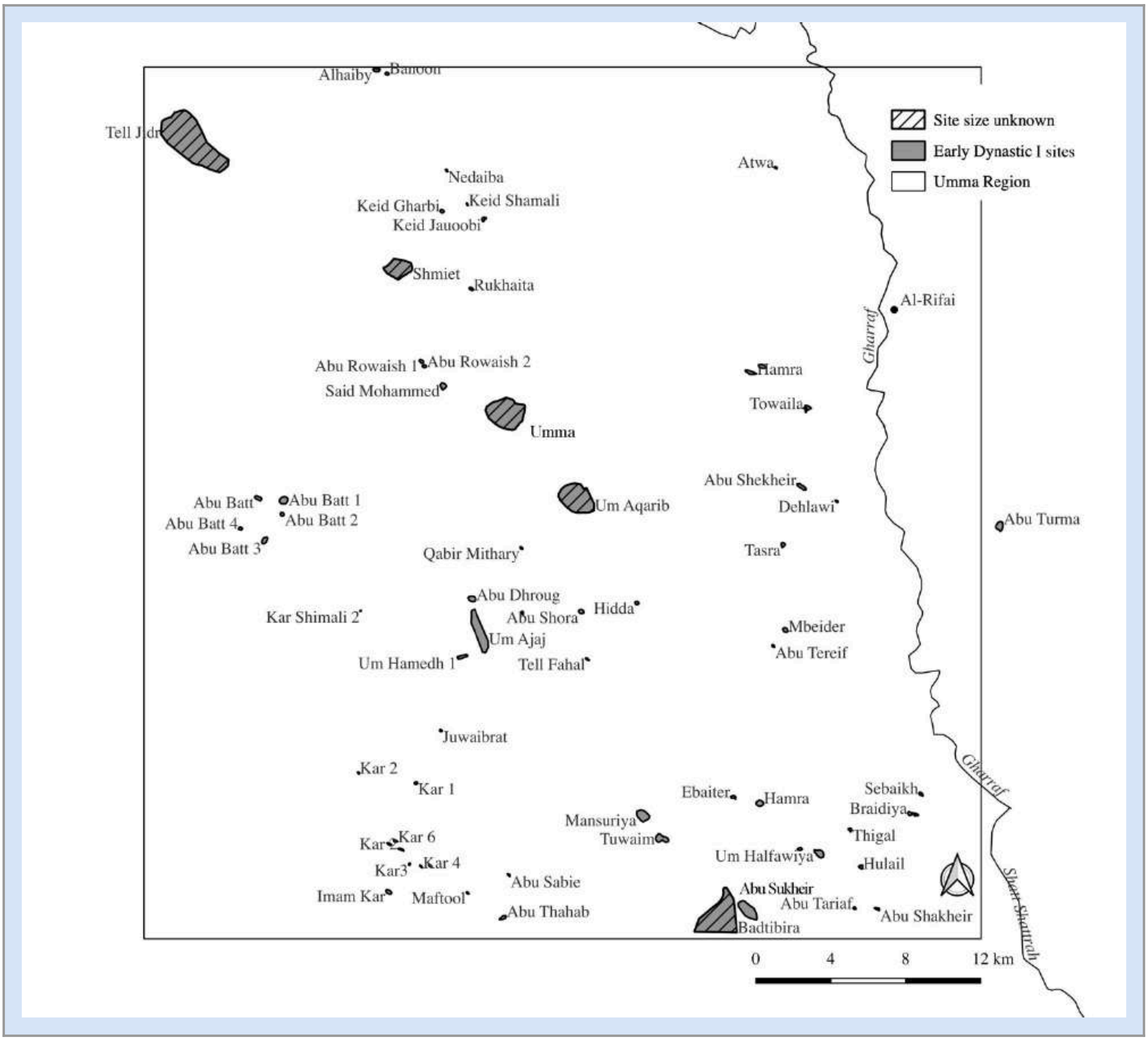

Figure 7: The Early Dynastic I (2900–2700 BC) occupation of the Umma region

While the sites of this period are more evenly spread over the Umma region there are also some noticeable settlement clusters, as for example, Kar 1–5 surrounding the medium-sized settlement Imam Kar (5.7 ha), which has clear traces of a square shaped public building with towers (see figure 7). This site was situated along the Shatt al-Kar ("the canal that has been re-dug"), which carried irrigation water until the 1950s. The Shatt al-Kar was created by dredging a much older watercourse that may date as far back as the Early Dynastic period (2900–2350 BC) and probably connected once the settlement cluster Kar 1–5 and Imam Kar, then ran west of Um Ajaj and connected to the Abu Bott 1–4 settlement cluster. Another watercourse somewhat parallel to the Shatt al-Kar must have run east of Umma (Tell Jokha) and Umm Aqarib to supply the many new sites in that part of the region. Settlements of the Umma region did not yet show the typical pearl-on-a-string organization that becomes more pronounced in the Early Dynastic II/III (2600–2350 BC) and Akkadian period (2350-2120 BC). Rather, the sites tend to cluster, which may be an indication that those settlements relied on modified crevasse splays for irrigating agricultural fields, as discussed by Wilkinson, Rayne, and Jotheri (2015).

The extent of Umma (Tell Jokha) and Um Aqarib in the Early Dynastic I period (2900–2600 BC) remains somewhat unclear. Contrary to prior assumptions, there is now convincing evidence that parts of Umma (Tell Jokha) were already settled in the Early Dynastic period but experienced their greatest growth in the Ur III period (Almamori 2014, 2–3). Almamori (2014) further argues that the toponym Umma (Sum. GIŠ.KUŠU2.KI traditionally read as UmmaKI) in Early Dynastic royal inscriptions refers to the site of Um Aqarib and not to Tell Jokha (Umma) as has been traditionally assumed. He argues that Umm Aqarib is the Umma of the Early Dynastic period and the center of the fledgling city-state of Umma. Even though exact survey data does not yet exist, it can be assumed that Tell Jokha (Umma of the later periods) was probably dwarfed in size compared to Umm Aqarib, the Umma of the Early Dynastic period (2900–2350 BC). The other urban site in the Umma region during the Early Dynastic I period (2900–2600 BC) is Abu Sukheir with 65.5 ha.

The case of Um Aja with its 110.7 ha in size is less clear. There are no traces of any administrative or public buildings. Most remarkable are the multiple holloways that radiate out from the sites which suggest frequent movement of animals and people (see Ur 2002, 2003 for the case of Upper Mesopotamia). Um Ajaj and the surrounding subsidiary sites were in an area that still forms today a large depression, suggesting that it was once a large wetland. Raising animals in the midst of marshes requires high and dry ground which Um Ajaj could have provided, and given the many tracks leading from the site, it is possible that it functioned as a large animal production site. Cattle pens (Sum. e_2-tur_3) are mentioned in texts from the Umma archive of the Ur III period (2112–2004 BC) (Stępień 1996, 56–57) and thus far it remains unclear where such pens could have been located. A more intense investigation of the site itself should provide some insight into the function of the site.

9. Early Dynastic II/III Period (2600–2350 BC)

The Umma region experienced a drastic transformation into an urbanized landscape at the time when the Umma region developed into an important city state (see figure 8). The total number of sites still increases from 66 in Early Dynastic I period (2900–2600 BC) to 85 in the Early Dynastic II/III period (2600–2350 BC) (see figure 2, table 3). Five settlements (Abu Sukheir, Muhaliqiyat, Tell Shmiet, Um Ajaj, Um Aqarib) expanded and developed into small and large cities (< 50 ha). As the extent of the Early Dynastic II/III period occupation cannot be established with certainty for all sites, the total area covered by cities might be slightly inflated. Almamori (2014, 2), for example argues that Umma (Tell Jokha) was only a small settlement during the Early Dynastic II/III period. Um Aqarib, on the other hand, grew to encompass an area of 252 ha and was the main and largest urban center in the Umma region. As the site was abandoned at the end of the Early Dynastic II/III period, the size of the site during that period is certain.

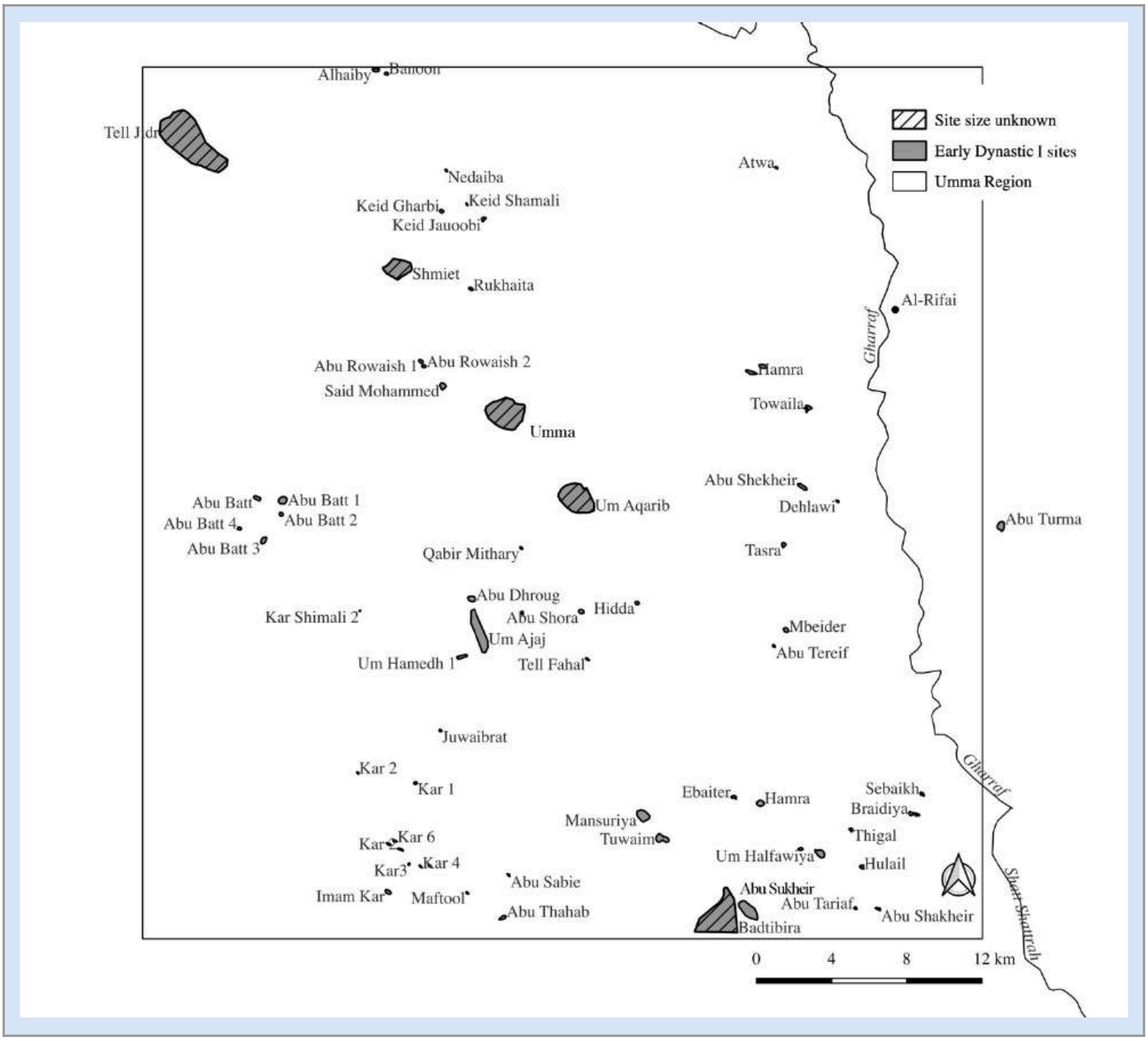

Figure 8: The Early Dynastic II/III period (2600–2350 BC) occupation of the Umma region

As has been mentioned earlier, the data from the Umma region suggest that the growth of Early Dynastic cities did not happen at the expense of rural settlements. There is still a slight increase in the number of hamlets/special purpose sites (> 2 ha, 4 more sites) and small villages (2–4 ha, 6 more sites). Small towns (9–16 ha) nearly doubled from six in Early Dynastic I period (2900–2600 BC) to eleven in Early Dynastic II/III period (see figure 2 and table 3). While Adams's (1981, 138–141; Adams and Nissen 1972, 11–12, 17) data suggest an urban growth based on rural migration and the total collapse of rural settlements, the data from the Umma region do not confirm such trends. To the contrary, the urban growth in the Umma region appears to have been based on population growth that affected both urban as well as rural settlements (see figure 3 and table 4). It is conceivable that it entailed, next to natural growth, an influx of people into the region from elsewhere. Nevertheless, the data from the Umma region also show that there was a very noticeable trend towards urbanization with the number and size of cities expanding and 75% of its population living in urban centers (see figure 3 and table 4). However, it needs to be emphasized that the data also show that the rural population increased, even though only slightly. The rural depopulation has often been interpreted as a reaction to lasting conflicts between city-states, which is particularly well documented for the city states of Umma and Lagash (Cooper 1983). However, the finding for the Umma region cast some doubt onto this interpretation. It is equally possible that the rural depopulation documented by Adams (1981) may not before had been as drastic and more the result of the biases introduced by his surveying approach which tended to overlook small rural sites. As my survey was conducted over a longer time frame, with frequent visits to the region and having greater access to local knowledge about the existence and location of sites, I argue that resurveying the Nippur-Uruk region could yield similar results. While the trend towards greater urbanization is confirmed in both regions, understanding the underlying cause of that development might have to be re-investigated. The increase in small rural settlements would also suggest that agricultural areas are expanding, potentially to support those new urban centers. There appears to be an additional and probable canals system between Tell Shmiet, Muraibe and Muhaliqiyat, which would speak for that.

10. Akkadian Period (2350–2120 BC)

The number of settlements dating to the following Akkadian period drops slightly from 85 to 70 and close to 15 settlements got abandoned, including the major center of Um Aqarib, however, probably in favor of a growing Tell Jokha, the post-Early Dynastic Umma (see figure 2, 9 and table 3). Umma (Tell Jokha) becomes an important regional center in the kingdom of Akkad and the region was one of its main provinces (Foster 1982; Frayne 1993, 261–262), even though the size of Umma itself during the Akkadian period remains unknown. In addition, the apparent shift of the urban population from Um Aqarib to Tell Jokha remains unclear; however, could have been connected to changes in the hydrology of the Umma region (see figure 9). The settlement distribution suggests that there was a decline of small branching watercourses when compared to the Early Dynastic period (2900–2350 BC). Such a consolidation of water into a single major channel appears to have happened along the Shatt al-Kar that most likely once here connected the settlement cluster Kar 1–5 and Imam Kar with the Abu Bott 1–4 settlement cluster and ran west of Um Ajaj. The watercourse system running from Shmid to Tell Muhaliqiyat which became prominent in the Early Dynastic II/III period, may have declined as well. This observation is congruent with developments observed by Adams (1981, 19, 165–170) in the Nippur-Uruk region as well as with the results of the QADIS survey (Marchetti et al. 2019, 224). Beginning with the Early Dynastic period the configuration of river systems changed from an anastomosing to a dendritic one, which caused the former web of smaller interconnected tributaries that ran across the landscape to dry up and led to a consolidation of the river's water in a few single channels. It is possible that the abandonment of Um Aqarib at the end of the Early Dynastic II/III period (2600–2350 BC) is related to those environmental changes. Similarly, the abandonment of Um Ajaj, another major site of the Early Dynastic period might be related. However, many smaller rural sites were affected as well. While some of the smaller settlement clusters, such as Kar 1–5 survived, many smaller sites, particularly those clustering around Badtibira (Tell al-Medain) declined.

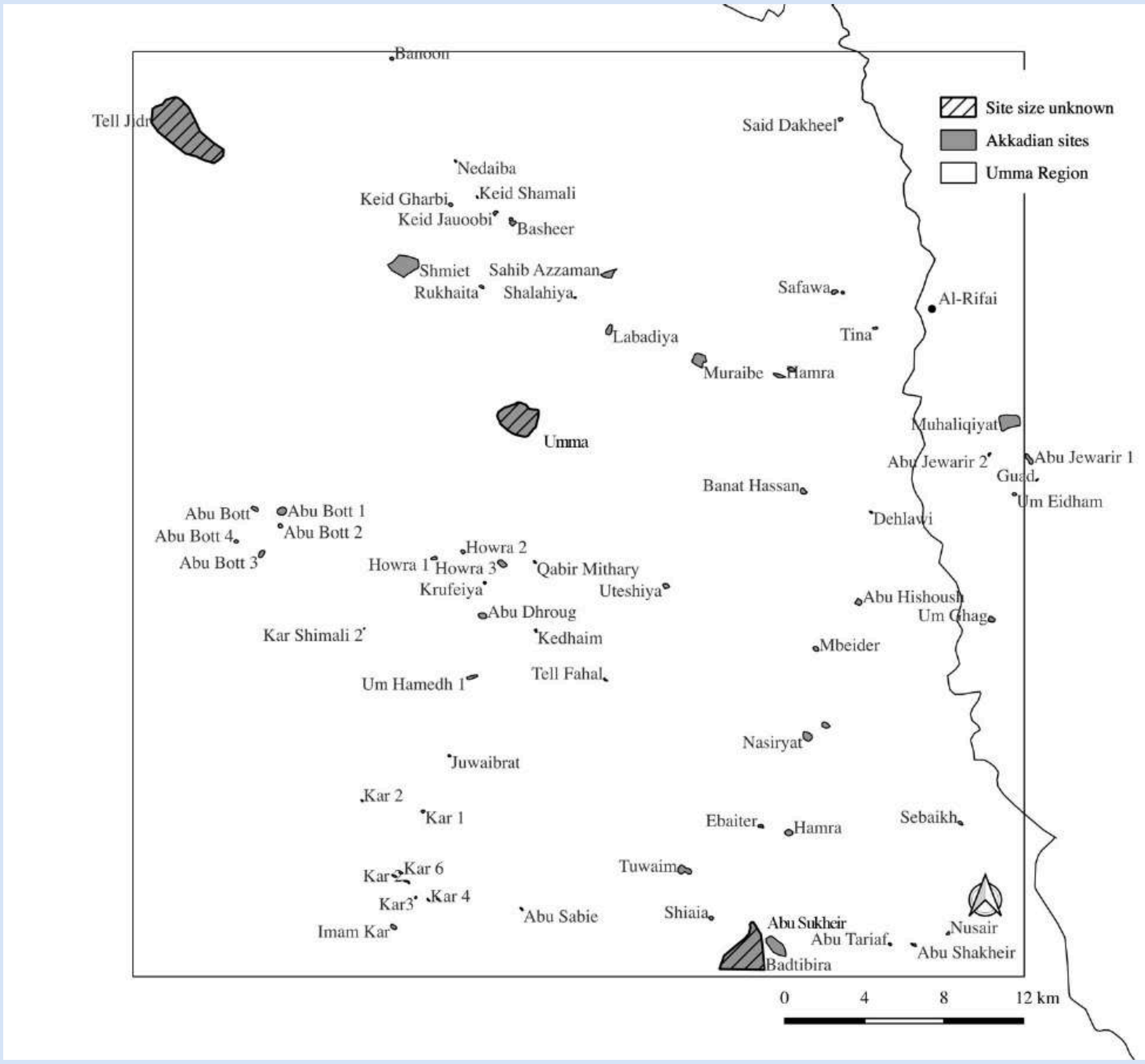

Figure 9: The Akkadian period (2350–2120 BC) occupation of the Umma region.

11. Ur III/Isin-Larsa (2112–1850 BC)

The Umma region reaches its peak both in total occupied area and number of settlements during the Ur III/Isin-Larsa period (see figure 2, 3, 10 and table 3 and 4). A sharp increase in the number of sites was also documented in the QADIS survey area, even though the number of sites does not surpass the number of sites of the Uruk period (Marchetti et al. 2019, 224). While these two periods are historically distinct, it is thus far not possible to distinguish between them based on diagnostic pottery fragments, which are the main data source in archaeological surveys (Adams 1981, 170–174; Adams and Nissen 1972, 37). The number of sites rose from 70 in the Akkadian period to 108 sites in the Ur III/Isin-Larsa period, which is a 54% increase in total number of sites (see table 3 and 4). Adams (1981, 151) only document nineteen sites dating to the Ur III/Isin-Larsa period. Written sources indicate, however, that there were at least 110, and possibly 158, settlements in the Umma province in the Ur III period (Steinkeller 2007). In addition, based on his studies, Steinkeller (2007, 196) further argued that most settlements fell into the hamlet/small village category and the settled landscape really represented a “rural continuum”. This conclusion can partially be confirmed by archaeological data. For one, 58% of the settlements dating to the Ur III/Isin-Larsa period fall into the hamlet

(>2 ha)/small village (2-4 ha) category and the increase in the number of settlements was most pronounced in those (see figure 2, 3 and table 3,4). I can also confirm that the Umma region was densely settled during the Ur III and Isin-Larsa periods, an observation that was also made by Adams (1981, 150–151) in the central alluvial plain but also by Wright (1981, 129–130) for the Ur-Eridu region. Adams (1981, 151) notes, however, that the increase is somewhat exaggerated given the conflation of Ur III and Isian-Larsa period.

Some of the larger sites, that had already been inhabited during the Akkadian period (2350–2120 BC) such as Umma (Tell Jokha) and Tell Shmid and Tell Muhaliqiyat, probably grew to urban size in the Ur III and Isin-Larsa period (see figure 10). However, there are also a number of new foundations of towns and cities, such as Labadiya (13 ha), Nasiryat (29 ha) and Zabalam (Tell Bizikh, 125 ha). Nasiryat must have been an important town due to its size and the density of the surface record and is a strong candidate to represent the ancient town of Garshana. Written documents indicate that it was located on the eastern border of Umma with the neighboring province of Girsu (Steinkeller 2011). Tell Muhaliqiyat (Steinkeller 2001, 34–35) was most likely located at the former border between the Umma and Girsu provinces and may have been the ancient site of Apishal (see figure 10). The site extends over 73 ha with a wall that encircles what appears to be a square-shaped public building. The wall got exposed by looting but also by extracting construction materials for mud bricks used in the surrounding villages. Administrative documents from the Ur III period (2112–2004 BC) locate a major barrage at the Tigris at Apishal (Rost 2019, 37–39). According to an in-depth study of historical records and the available archaeological record, Steinkeller (2001) argued (and this is now widely accepted) that the remnants of an ancient levee running Jidr, Shmid, Bizikh, Muhalqiyat, was the course of the ancient Tigris in the late third and early second millennium BC. The settlement data that I collected suggest that the ancient Tigris was splitting into several branches at the location of Tell *Muhaliqiyat* with one running south-east towards Girsu (Tello), Lagash (Al-Hiba) and Nina (Tell Zurghul), one running north-south towards Badtibira (Tell al-Medain) and a third running north-southwest towards Nasiriyat. Assuming that Tell Muhaliqyat was indeed Apishal as suggested by Steinkeller (2001, 34–35) this split of the ancient Tigris into several branches might have been facilitated by the barrage (Sum. kun-zi-da) at Apishal. Located at such a crucial intersection, I argue that Apishal played a pivotal role in controlling river-born transportation, internal trade, and water distribution (see also Al-Dafar Al-Hamdani 2015b, 146–157).

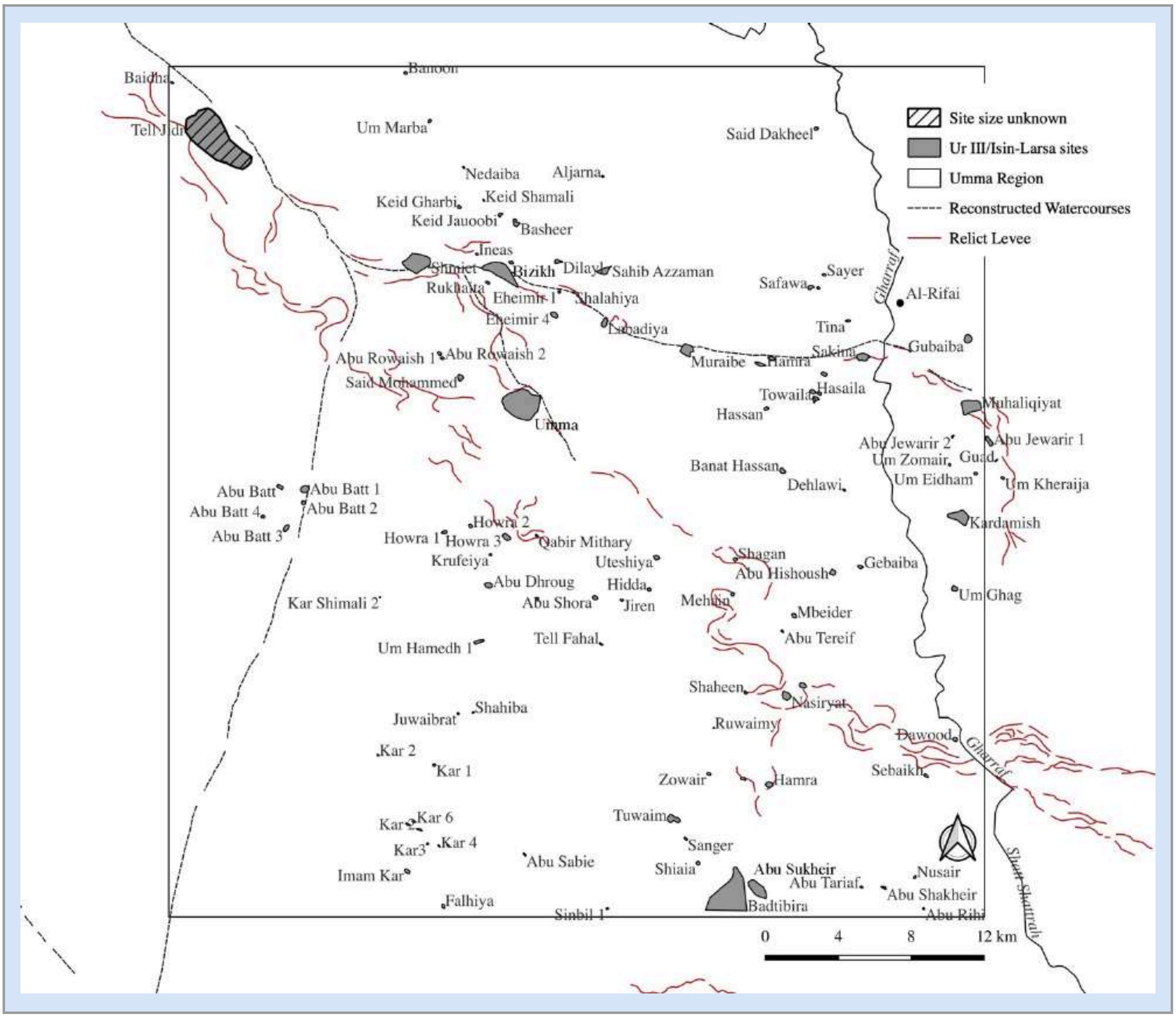

Figure 10: The Ur III – Isin-Larsa Period (2112–1850 BC) occupation of the Umma region.

12. Old Babylonian (1850–1595 BC)

The Umma region experiences a notable decline in the number of settlements in the following Old Babylonian period, which dropped from 108 in the Ur III/Isin Lara period to 62 sites in the Old Babylonian period (see figure 2, 3, 11 and table 3,4). It appears that Old Babylonian settlements concentrate along the watercourse that ran from Tell Jidr via Shmiet and Zabalam (Tell Bizikh) eastwards to Tell Muhaliqiyat and Girsu (Tello). The channel/watercourse running north south connecting Umma with Badtibira (Tell al-Medain) seems to have gone out of use. However, Badtibira was occupied in the Old Babylonian period and must have received water from a different source. The decline in settlements in the Umma region is probably related to the political and economic crises during the reign of Hammurabi's son Samsuiluna, which had a particular adverse effect on the Nippur-Uruk region (Boivin 2022, 615–625; Ur 2013, 145–147). The political turmoil and economic distress were most likely related to water shortages that had affected many very prominent southern Mesopotamian cities, such as Nippur, Isin (Ishan al-Bahriyat), Larsa and Ur and Uruk for some time.

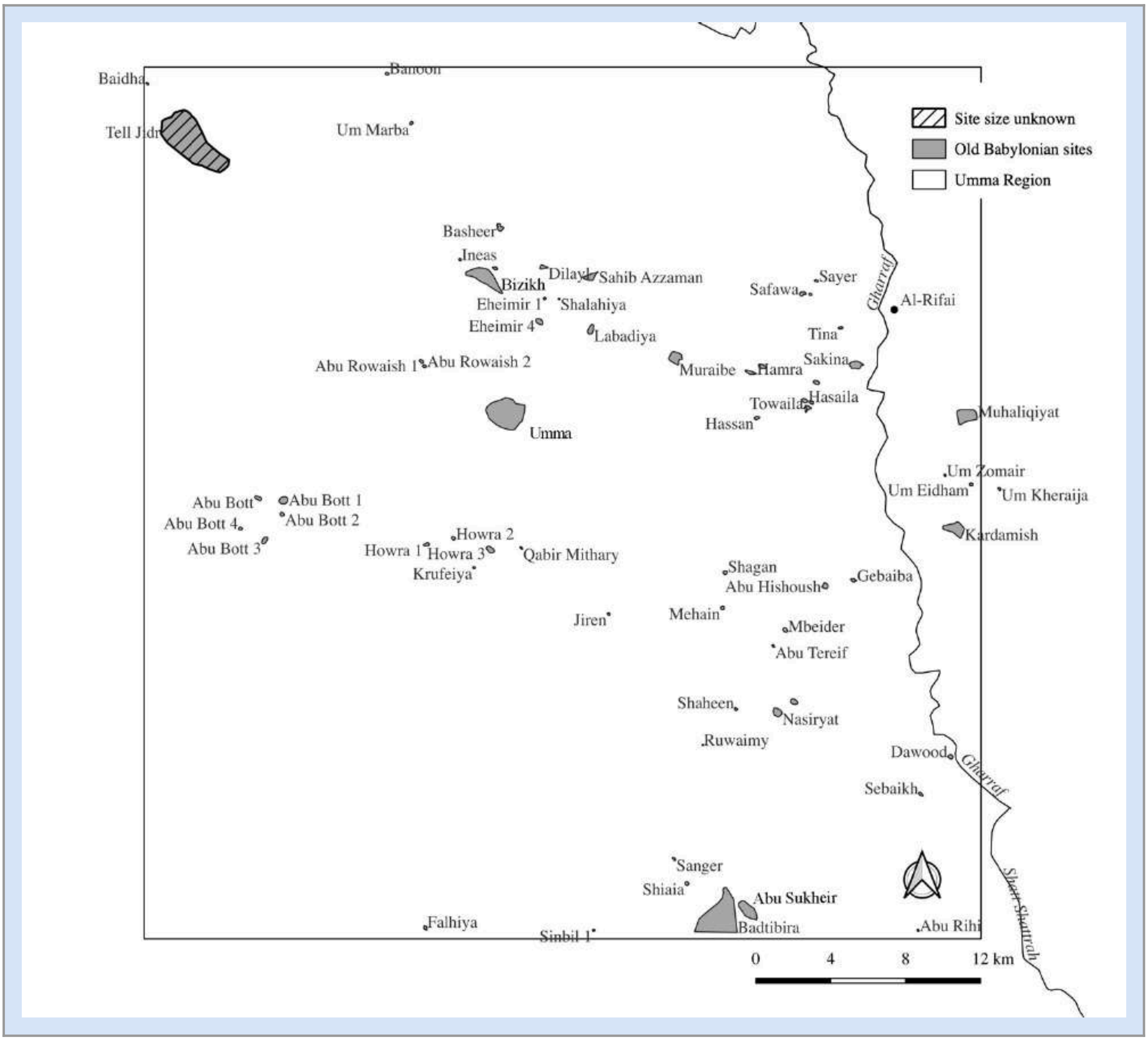

Figure 11: The Old Babylonian period (1850–1595 BC) occupation of the Umma region

A period of low precipitation rates in the Euphrates and Tigris catchment in the Taurus mountains, between the third millennium and the fifth century B.C. drastically reduced the water volume that both rivers carried. Lower water volumes reduced the flow velocities in the rivers and increased the deposition of silt in the river's main channel which over time became narrower and caused many major and minor river branches to dry up. A constricted main river channel not only caused water shortages, particularly further downstream, but also increased the risk of devastating floods. Annual variations in precipitation in the catchment area could cause an influx of water that the constricted river channels could no longer contain. The construction of dikes to protect cities such as Nippur, Sippar (Abu Habbah), Amnanum (Tell ed-Der), Uruk (Warka) suggests that such floods were frequent in the first half of the second millennium BC (Cole and Gasche 1999, 7–11).

Written sources also show that many Old Babylonian rulers, including Hammurabi, got actively involved in trying to maintain or restore water flow to those important cities, as well as protecting them from floods. All those measures, however, were only temporary solutions (Rost 2017, 12–14). Hammurabi's son Samsuiluna faced many rebellions from the south, which may have been related to those environmental stressors and ultimately led to the disintegration of his kingdom. His inability to maintain control over the south may have been equally related to the expansion of marshland in this period. Frequent floodings as well as high water levels in the Persian Gulf may have led to the south becoming increasingly swampy and inaccessible for the land-bound armies that Samsuiluna had at his disposal. The independence of the south from its Babylonian overlords ultimately allowed for the Sealand Dynasty to take control of much of the south (Al-Dafar Al-Hamdani 2015b, 2020; Boivin 2022). The Ur-Eridu region, as a result flourished (Wright 1981, 330–331; Ur 2013, 147), while the Nippur-Uruk region became increasingly de-urbanized and rural (Adams 1981, 165).

The shift towards a more rural and less urbanized settled landscape occurred in the Umma region slightly later with the beginning of the Kassite period (1475–1155 BC). Even though there is a noticeable decrease in the number of settlements, the total occupied area of cities even increases (see figure 2, 3 and table 3, 4). Major urban centers such as Umma (Tell Jokha), Badtibira (Tell al-Medain), and Zabalam (Tell Bizikh) show extensive occupation and may have been inhabited for most of the Old Babylonian period (see also Adams and Nissen 1972, 38). This is also true for smaller urban sites, such as Kardamish, Abu Suhkeir and Muhaliqiyat. Interestingly, the reduction in sites occurred primarily among hamlets, small and large villages, where the number in each of those categories fell by nearly half from the preceding Ur III/Isin-Larsa period (see table 3). This could be related to the desiccation of smaller tributaries of the ancient Tigris River. Most of the larger sites were located along what is now believed to be the ancient course of the Tigris River that ran from Tell Jidr via Tell Shmiet and Zabalam (Tell Bizikh) eastwards to Muahliqyat and Girsu (Steinkeller 2001).

13. Kassite Period (1475–1155 BC)

The Umma region experiences a dramatic decline in the total settled area and a widespread depopulation of the region (see figure 2, 3, 12 and table 4). The number of known sites with Kassite occupation drops to 14 from 62 in the Old Babylonian period (see table 3). Most sites fall into the rural category (hamlets, small and large villages) and only one urban site survives – Abu Suhair (65 ha) – most likely, however, on a much smaller scale (see figure 11). Tell Jidr seems to have some Kassite occupation, but it remains unclear how large (see also Adams and Nissen 1972, 40, Fig. 18, 19). The widespread abandonment is most likely related to a major shift of the Tigris to the east, leaving the prior densely inhabited areas in Umma without water. A similar shift to the west occurred along the Euphrates, depriving many areas of water and led to the consecutive abandonment of major cities such as Larsa, Ur and Uruk (Adams and Nissen 1972, 39–41). Much of the previously cultivated land went out of use and many inhabitants of the south may have moved north closer to the political center, such as Babylon. It is likely that a large portion of the population may have returned to a nomadic lifestyle based on animal herding (Malko 2014, 97–102; Sassmannhausen 1999). Most Kassite settlements were concentrated around Babylonia, which was the political center of the Kassite Kingdom and most of the south was probably without water or had turned into a swamp. There is an occupation hiatus between the end of the Kassite period and the beginning of Neo-Babylonian period (620–539 BC).

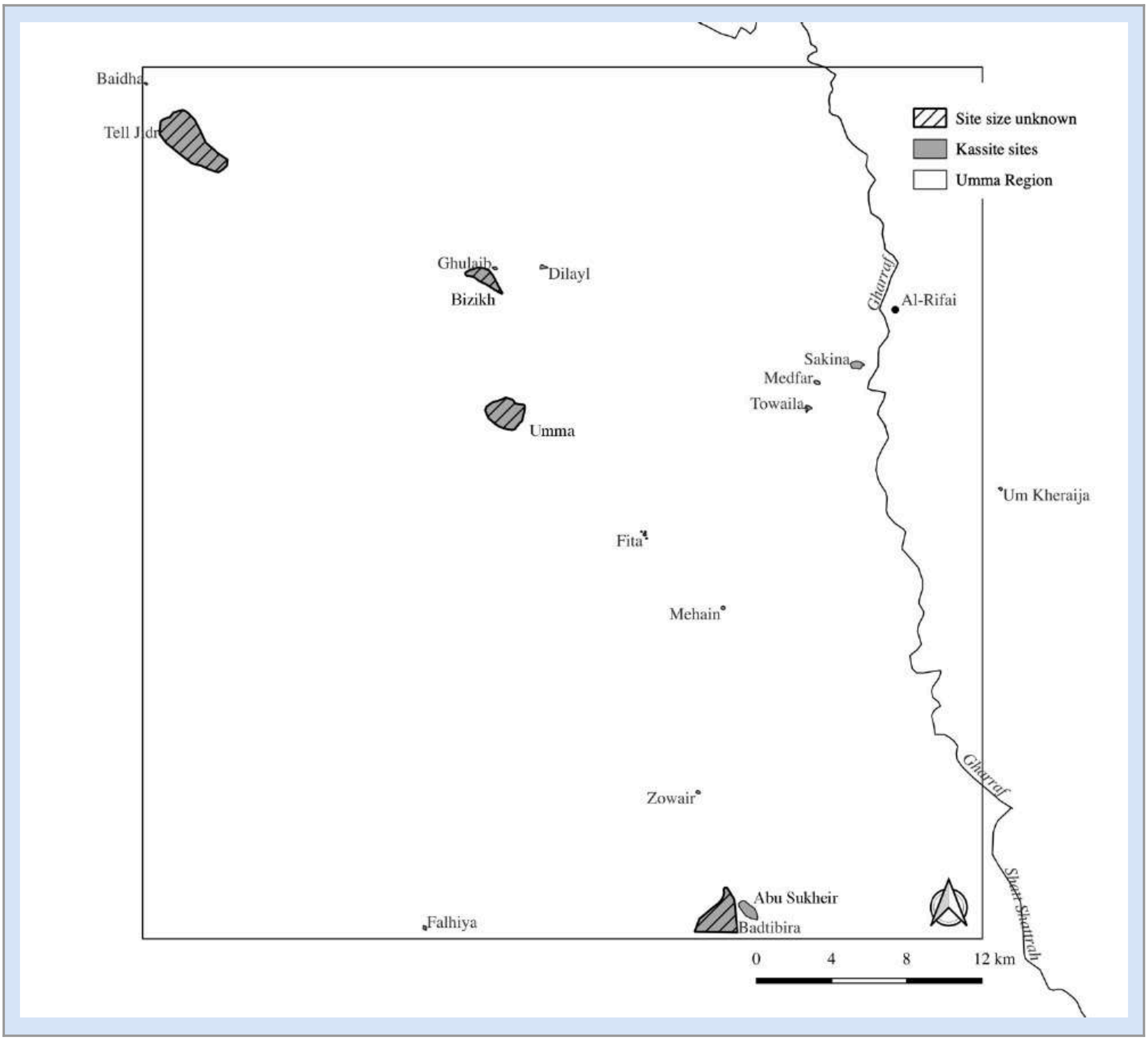

Figure 12: The Kassite Period (1475–1155 BC) occupation of the Umma region.

14. Neo-Babylonian Period (620–539 BC)

The Neo-Babylonian period marks the beginning of a revival of southern Mesopotamia which from thereon is firmly integrated into empires, who controlled far greater territories than any of the kingdoms prior. Between the Neo-Babylonian and the Sasanian period (AD 224–651) Southern Mesopotamia experienced a steady and marked population growth, a second apogee in urbanization and an impressive extension of agricultural regimes that entailed large-scale interconnected and state-sponsored irrigation systems that ultimately encompassed the entire alluvial plain (Adams 1981, 177–179; Adams and Nissen 1972, 57; Jursa 2010, 437–443). Adams (1981, 177–178) attributes the population increase to a factor of five over the course of 500–700 years (between the Middle Babylonian (1158–1026 BC) and the Seleucid/Parthian period [312 BC – AD 224]) to conditions of political stability and economic prosperity fostered by imperial dynasties. However, given the superior agricultural productivity of the alluvial plain, southern Mesopotamia functioned as a granary for the Neo-Babylonian as well as succeeding empires, and its population had to bear heavy tax burdens with far less influence on the political institutions that ruled them (Adams 1981, 175). Adams (1981, 175) even suggests that, controlling the peasantry was vital to maintain the flow of grain revenues to fund the military and political ambitions of those empires. The emphasis on rural settlement in the resettlement of the southern alluvium, including the Umma region, might have been quite deliberate.

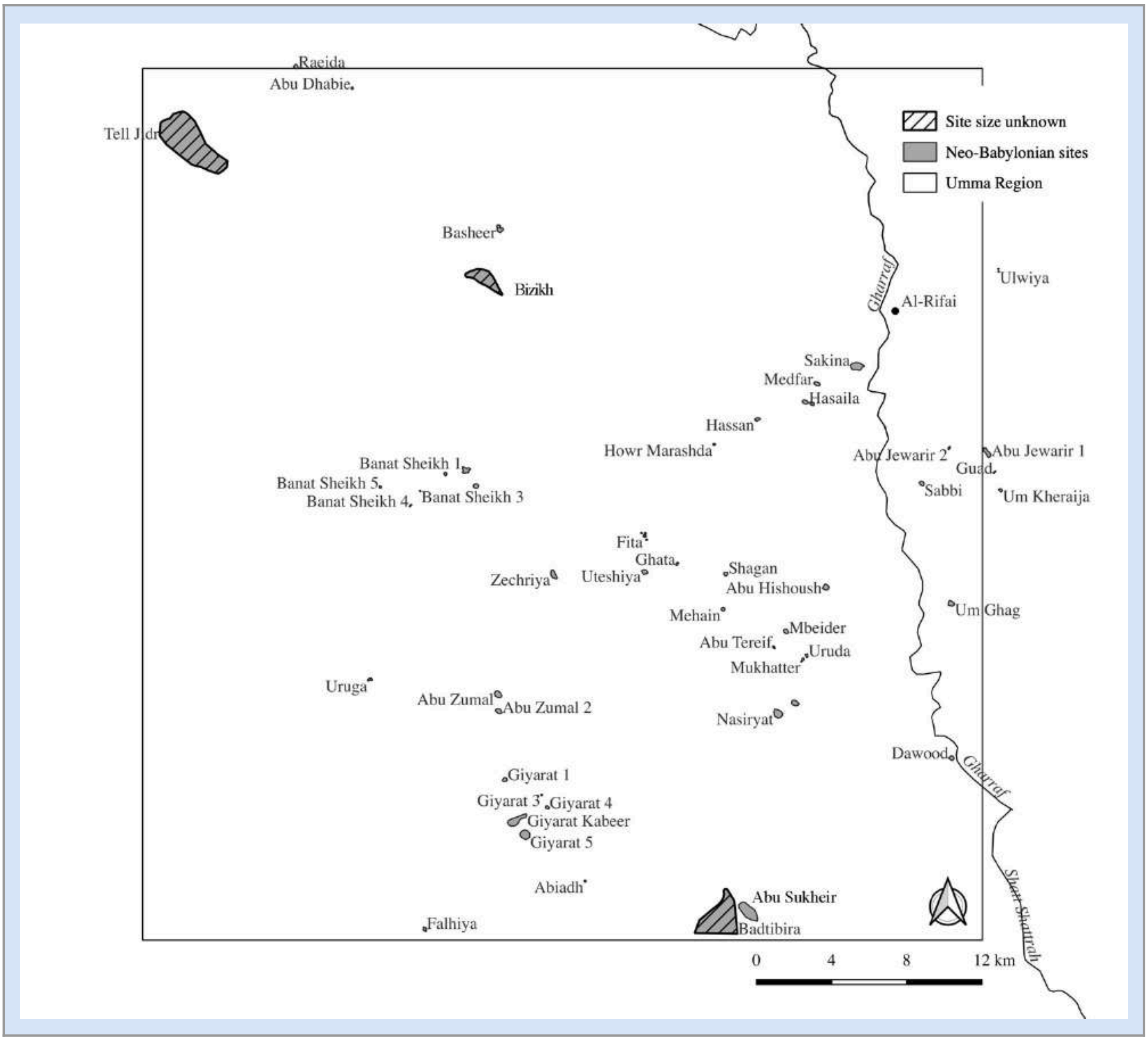

Figure 13: The Neo-Babylonian Period (620–539 BC) occupation of the Umma region

There is a noticeable increase in the number of sites in the Umma region dating to the Neo-Babylonian period which rose from 14 in the Kassite period (1475–1155 BC) to 53 settlements. The sudden increase in the number of settlements has also been observed by Adams and Nissen (1972, 54) farther to the west and can most likely be attributed to the forceful resettlement of deported populations from other regions. However, in contrast to earlier periods, the settlement pattern that emerges is very rural, with 72% of the sites falling into the hamlet (>2 ha), small (2–4 ha) and large village (5-8 ha) categories (see figure 2, 3, 13 and table 3,4). There are several small (6), regular (3) and large towns (2) but only one relatively small city (Abu Suhair with 65 ha). The Neo-Babylonian occupation of the former large urban sites, such as Badtibira (Tell al-Medain) and Tell Bizikh (Zabalam) were small and concentrated in the case of Tell Bizikh in the northern part and in the case of Badtibira on top of the main mount. The interest of the Neo-Babylonian kings in restoring southern urban centers was very selective. Nabonidus, for example restored the ziggurats at Ur and Eridu, the Ebabbar at Larsa and the akītu-tempel at Uruk. At Uruk he even rebuilt the “Royal Canal” (Akk. nâr šarri) which indicates the development of agricultural land. However, the rest of the south remained undeveloped throughout the Neo-Babylonian period, particularly the Umma region (Beaulieu 1989, 17, 42, 210).

Moreover, the configuration of the sites differs greatly from the heavily urbanized phases of the Umma region in the Early Dynastic II/III (2600–2350 BC) and Ur III/Isin-Larsa periods (2112–1850 BC). Once more, the settlement pattern is dominated by clusters of small settlements (e.g., Banat Sheikh 1–5, Giyarat 1–5, etc., see figure 13). This settlement configuration was first observed in the Jemdet Nasr (3100–2900 BC) and Early Dynastic I period (2900–2600 BC) and could be related to a form of irrigation that relied on the use and modification of crevasse splays (Wilkinson, Rayne, and Jotheri 2015). It is certain, however, that the hydrology of the Umma region had drastically changed from the earlier periods, particularly with the ancient course of the Tigris River having shifted further to the east sometime around the end of the Old Babylonian period (1850–1595 BC). Thus, it is not entirely clear what this settlement pattern tells us about the hydrology and land use in the Umma region during the Neo-Babylonian period.

15. Achaemenid Period (539–330 BC)

After the conquest of Babylon by Cyrus the Great in 539 BC, southern Mesopotamia was firmly integrated as the Babylonian satrap (province) into the Achaemenid Empire (539–330 BC). Even though it is certain that the developments (population increase, increased urbanization and expansion of agricultural and irrigation systems) continued and were enhanced during the Achaemenid period, the overall settlement history of the alluvial plain is not well understood. At the time when Adams (1981) conducted his survey, the pottery chronology of the Neo-Babylonian (620–539 BC), Achaemenid (539–330 BC) and Seleucid periods (312 BC–AD 224) was poorly defined. Adams (1981, 176) himself notes that any settlement pattern emerging from the surface collection of individual sites are highly debatable. Thus, Adams (1981, 177–190) ends up lumping the Neo-Babylonian and Achaemenid records and hence changes in the overall settlement pattern between these two periods cannot be described. The pottery chronology of these periods has been considerably refined since, particularly with new excavation underway in the south of Mesopotamia (Di Michele forthcoming). However, it will require resurveying much of Adams’s survey area to improve the resolution of the settlement history of the alluvial plain.

Most of the textual evidence points to continuity at least for the beginning of Achaemenid rule over southern Mesopotamia which may also be true of the settlement pattern of the alluvial plain. According to Potts (2021, 19–22) it is often assumed that Achaemenid rulers were actively involved in the expansion and intensification of the agricultural production in various regions of their empire by investing in the construction of canals or hydraulic devices. However, it is more likely that the Achaemenid rulers inherited a highly efficient and very productive agricultural and irrigation infrastructure from the preceding Neo-Babylonian Empire whose administration seems to show at first, a remarkable continuity until the rebellion of the Babylonian elite in 484 BC (Kleber 2021, 909; Pirngruber 2017, 27; Jursa 2007).

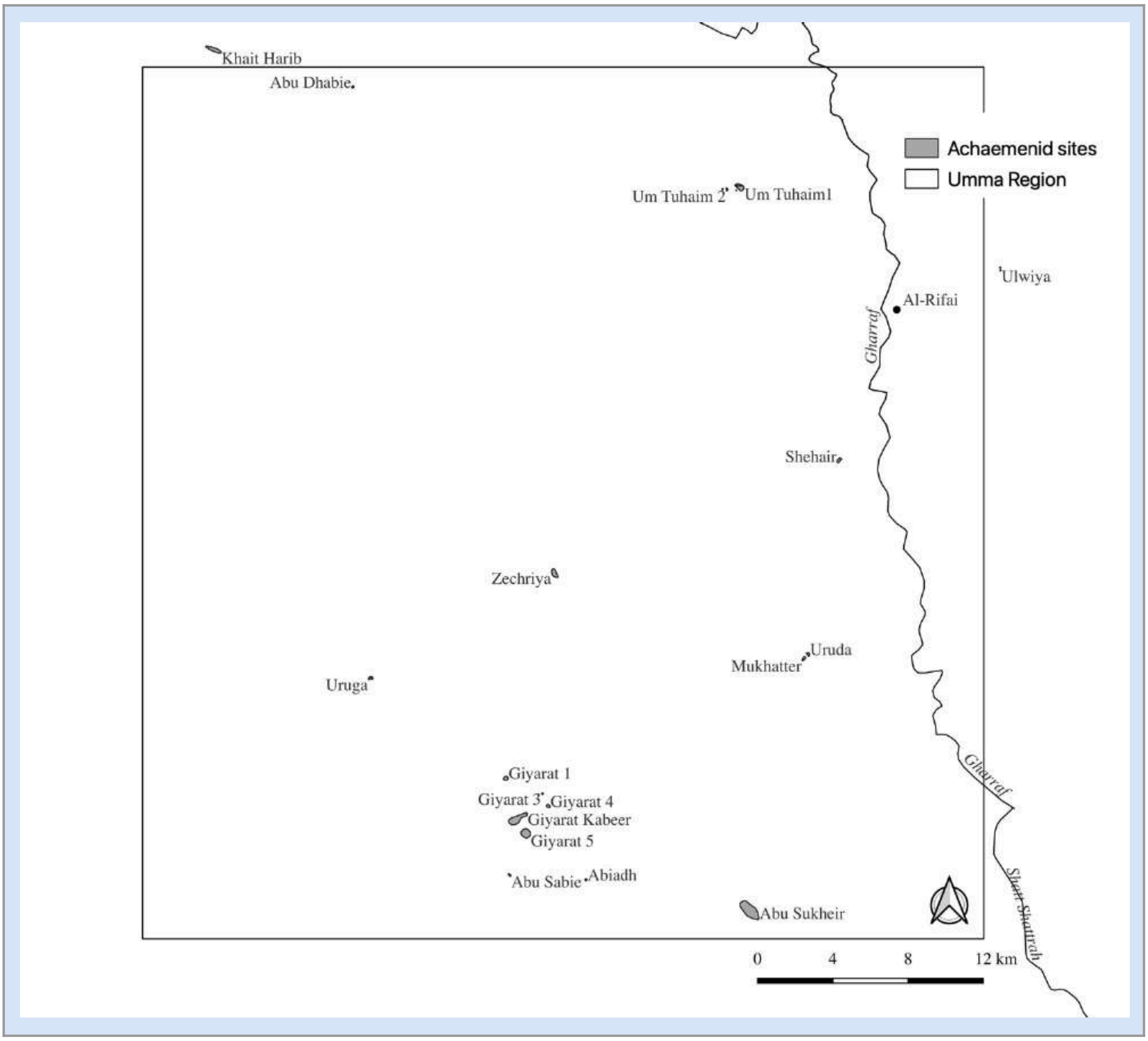

Figure 14: The Achaemenid Period (539–330 BC) occupation of the Umma region.

Achaemenid administrators in charge of the Babylonian satrap supervised the upkeep as well as the expansion of irrigation systems, however, the cost and labor burden were divided among private landowners and temple households who most likely also managed local project without the investment or supervision of the imperial administration (Potts 2021, 21, see also Adams 1981, 186). Thus, it is probably fair to say that the settlement pattern did not drastically change when compared to the Sasanian period, which was a in many regards very different from earlier periods.

The Umma region surprisingly shows a decline in the Achaemenid period with more than half of the sites being abandoned (see figure 14). The greatest loss is in the categories of small rural settlements. The number of hamlets (> 2 ha) declined from 12 to 5 and a loss of similar magnitude (14 to 6) can be observed in small villages (2–4 ha). Most affected, however, are large villages which shrink from 12 in Neo-Babylonian-times (620–539 BC) to only 3 in the Achaemenid period (539–330 BC, see figure 2, 3 and table 3, 4). According to Jursa (2010, 437–443); state-sponsored improvements to the irrigation infrastructure that increased agricultural productivity were varied. The northern part of the southern alluvial, in the Sippar region, investments in the irrigation infrastructure were greatest while in the very south around Uruk and Nippur they were lower. Moreover, these impressions are based on archives kept by entrepreneurs, who managed large tracts of land. The lack of comparable data from other regions, such as the Umma region, does not allow for any generalization (Adams 1981, 186). In addition, Jursa (2010, 406) notes that Nippur in the Neo-Babylonian and early Achaemenid periods appears to have been somewhat isolated and detached from major communication lines along the Euphrates and may not have had as vibrant an economy as other regions in southern Mesopotamia. Thus, the Umma region may have been even more isolated and therefore more prone to water shortages that cause a decline of the region.

16. Seleucid-Parthian (312 BC – AD 224)

Adams (1981, 179) observed a threefold increase in the number of settlements from the Neo-Babylonian period (620–539 BC) to the Seleucid-Parthian period and a ninefold increase in urban sites (above 10 ha in size) most of which are new foundations, pointing to "state-initiated schemes of urbanization and agricultural expansion" (Adams 1981, 179). Based on the data available to him, Adams and Nissen (1972, 58–59) noted that, except for Uruk and Jidr (675 ha), Parthian centers in southern Mesopotamia were not comparable in size and importance to cities of earlier periods, such as Umma, Zabalam (Bizikh), Badtibira and Larsa. It appears that southern Mesopotamian during the Seleucid-Parthian period was far less urbanized and dominated by towns that were smaller and much more dispersed across the landscape. Moreover, small settlements do not appear to have been subordinated to towns and cities and their distribution across the landscape was driven more by the necessary proximity to fields (Adams and Nissen 1972, 58–59). Adams (1981, 200) further remarks that, given that southern Mesopotamia was at the very margins of a "loosely integrated empire", control may have fluctuated and that the stimulus for agricultural development and expansion of settled area may have been based on the initiative by Mesopotamians themselves rather than a result of imperial policies.

For the Umma region, the number of sites doubled from the preceding Achaemenid period (539–330 BC) and rose to 41 with most new foundations found in the small settlement categories of hamlets and small and large villages. However, the most remarkable increase is in the number of cities (< 50 ha), of which there are four now (see figure 2, 3, 15 and table 3, 4). Thus, the impression that Adams and Nissen (1972, 58–59) had based on their data cannot be confirmed for the Umma region. While it is true that the new urban centers are not of the same magnitude, both in size and probably in importance, as the earlier Sumerian cities, they are still numerous and substantial in size, Abu Al-Suhair [106 ha], Abu Suhair [65 ha] Kardamish [61 ha] and Farwa [45 ha]) particular Tell Jidr (675 ha). Furthermore, there are new foundations of several small settlements called Banat Sheikh 1-7 which appear in organization and proximity to be subordinated to Farwa. Thus, the impression that Adams (1972, 58–59) had of the region as having less of an integrated settlement hierarchy cannot be confirmed.

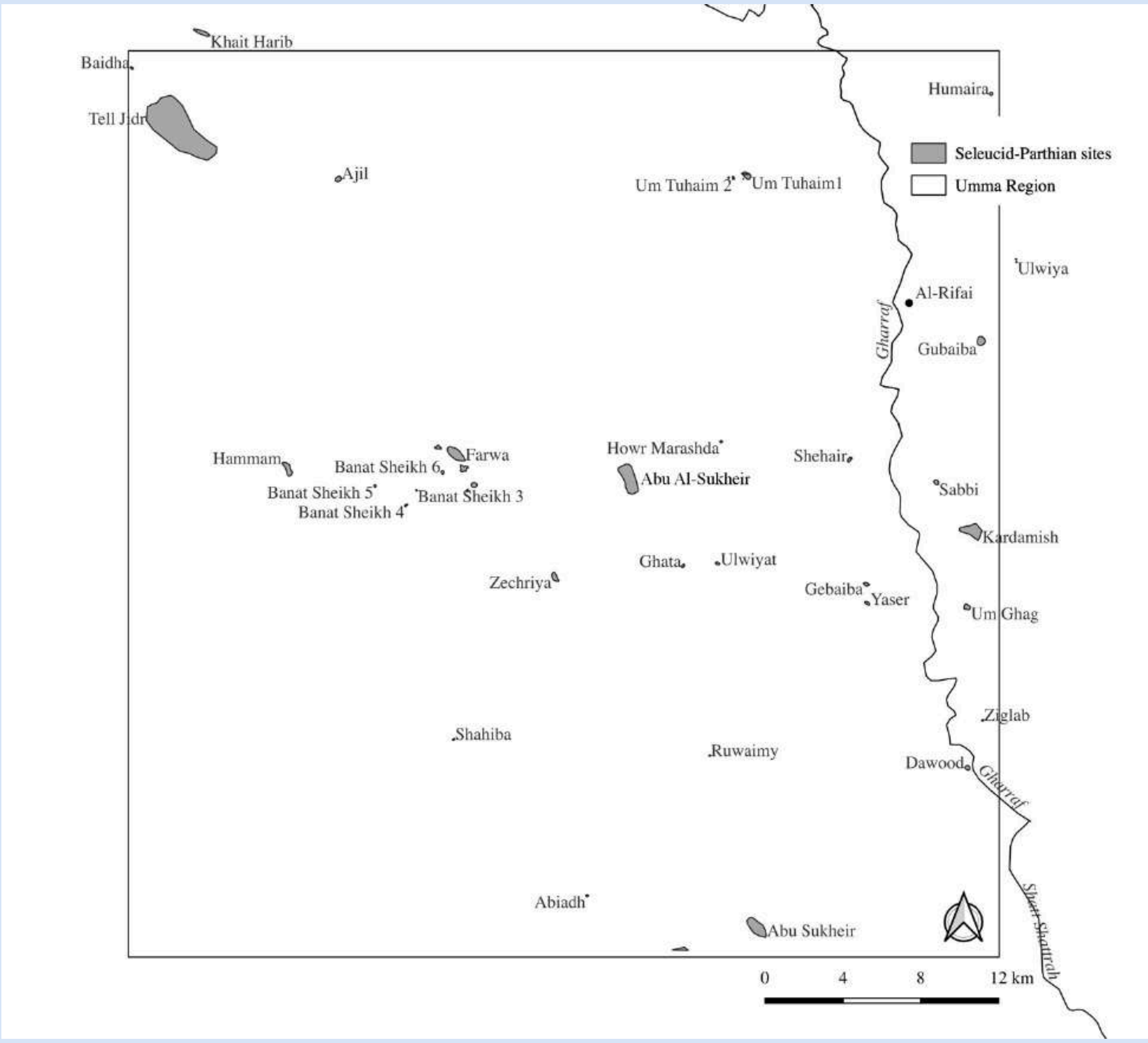

Figure 15: The Seleucid-Parthian Period (312 BC – AD 224) occupation of the Umma region.

The hydrology of southern Mesopotamia since the Neo-Babylonian period has been characterized by a more dendritic, increasingly artificially engineered water distribution system which encompassed most of the alluvial plain. In the Umma region there appears to be a canal running from Tell Jidr that passes Farwa and runs southeast towards Kardamish and down to Girsu, which had a Parthian occupation (see figure 15). While most of the Seleucid-Parthian settlements are concentrated along that trajectory indicating an ancient watercourse, there are several very dispersed smaller settlements, and it remains to be determined whether they relied on smaller tributary canals or may have constituted marsh settlements.

17. Sasanian Period (AD 224–651)

The Sasanian period in southern Iraq "was the apogee of premodern development of the region in every economic as well as demographic respect" (Adams 1981: 179). There was a 37 % increase in total occupied area from the Parthian to the Sasanian period and compared to the Ur III period (2112–2004 BC), which constitutes the first apogee in urbanization and population density, a 61% increase. Thus, there was an immense expansion of cultivated area and remarkable population growth during the Sasanian period. At the same time, the alluvial plain was far removed from important political centers and slightly less urbanized when compared to the Diyala that was near the regional political center of Ctesiphon. The very south of Mesopotamia experienced a demographic decline by late Parthian and early Sasanian periods as overirrigation had led to the spread of swampy conditions and, which ultimately caused the decline of Uruk (Adams 1981: 180).

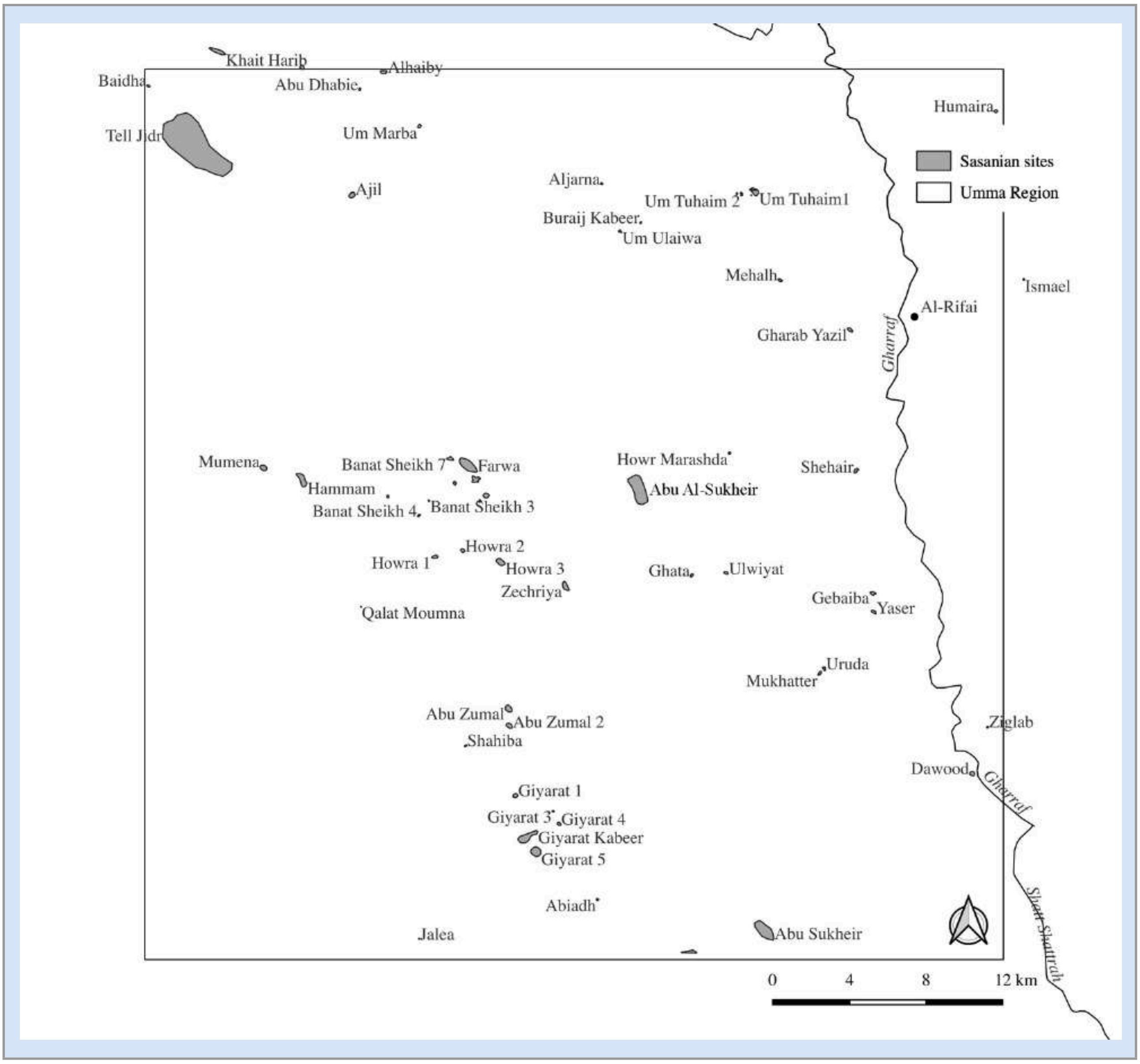

Figure 16: Sasanian Period (AD 224–651 BC) occupation of the Umma region.

The irrigation schemes built during the Sasanian period were very extensive and carefully planned and "involved a more ambitious reshaping of natural patterns of drainage than any previous ones" (Adams and Nissen 1972, 62). Canals ran perpendicular to the slope of the alluvial plain and created an interconnected system that covered a huge area in south-southeast direction. The Umma region certainly benefited from this state-sponsored investment in the agricultural and irrigation infrastructure and there was a considerable increase in the number of settlements (from 41 to 61) and total occupied area from the preceding Seleucid-Parthian period (see figure 2, 3, 16 and table 3, 4). The region is now known as the province of Kaskar and considerable investments were made into its agricultural development as the royal family turned it into a crown province that was granted to a member of the Sasanian royal household (Morony 1982, 32). The region might have received water from two major irrigation schemes with main canals branching off the Euphrates in the west and the lower, rerouted end of the Tigris to the east. However, the Tigris shifted its course shortly after the 5th century AD which led to disastrous floods and the spread of swamps. Some efforts were made to reclaim some agricultural ground however this was short-lived (Morony 1982, 32).

In contrast to the rest of the alluvial plain, the total area occupied during the Sasanian period does not surpass that of the Ur III/Isin-Larsa (2112–1850 BC), but instead comes in second (see figure 3 and table 4). The increase in the number of settlements was predominantly in the smallest site size categories: hamlets [< 2 ha] from 12 to 18, small villages [2–4 ha] 9 to 19. There was a minor increase in the number of large villages (5–8 ha) and small towns (9–16 ha) while the number of larger towns and cities remained stable compared to the preceding Seuleucid-Parthian period (see table 3 and figure 16). Thus, it appears that developing the agricultural potential of the Umma region did not lead to greater levels of urbanization but a denser occupation of the countryside.

However, the picture might be a bit more complicated since the distribution of the sites does not follow a predictable linear pattern along main or lower order canals (see figure 16). The settlements in the eastern part of the Umma region show a systematic and deliberate distribution following a line going from Jidr to Dawood in a northwest-southeast direction. Jidr was located at the tail end of a massive new canal system, along which major Sasanian cities were located, such as Nippur and Zibliyat to the north (see Adams 1981, 208, Fig. 45, 210, Morony 1982, 32). Its extent east of Jidr could not be reconstructed by Adams (1981), but it is possible that the smaller settlements in the eastern half of the Umma region were integrated into this large-scale irrigation scheme.

The distribution of sites in the southwestern part of the Umma region is dominated by clusters of small settlements (see figure 16). It is conceivable that those settlements may have exploited other sources of water such as crevasse splays or constituted settlements within extensive wetlands relying primarily on marsh resources. Arab historians of the 9th century AD mention a wetland or marsh area (Arabic Bāṭiḥa, plural Baṭāeiḥ, al-Balāthiry) south of the later Islamic city of Wāsiṭ (see Al-Dafar Al-Hamdani 2015, 81–86). The formation of this marsh area dates to the late Sasanian, specifically in the reign of Kavad I (488- 531 AD) and would support the notion that small clusters of settlements such as Giyarat 1–5 might be marsh settlements.

18. Islamic Period (AD 638–1924)

The early (AD 638–1000), middle (AD 1000–1517), and late Islamic period (AD 1517–1924) are summarized here since there is little change in the overall settlement pattern over the course of the Islamic period. Adams (1981, 184) observed further to the west a total reversal of the Sasanian maximum population growth and agricultural expansion. By the 11th–12th centuries AD the total occupied area in the central alluvial plain had diminished to 6% of what it had been half a millennium earlier. In addition, the settled landscape became exceedingly rural and dispersed and settlements of the size of 200ha + totally disappeared. Adams (1981, 185) attributes this demographic decline to the fact that the central alluvial plain had become marginalized as the Euphrates moved further to the west as well as migration out of the region, possibly close to the new urban centers of Basra, Kufa, and Wasit.

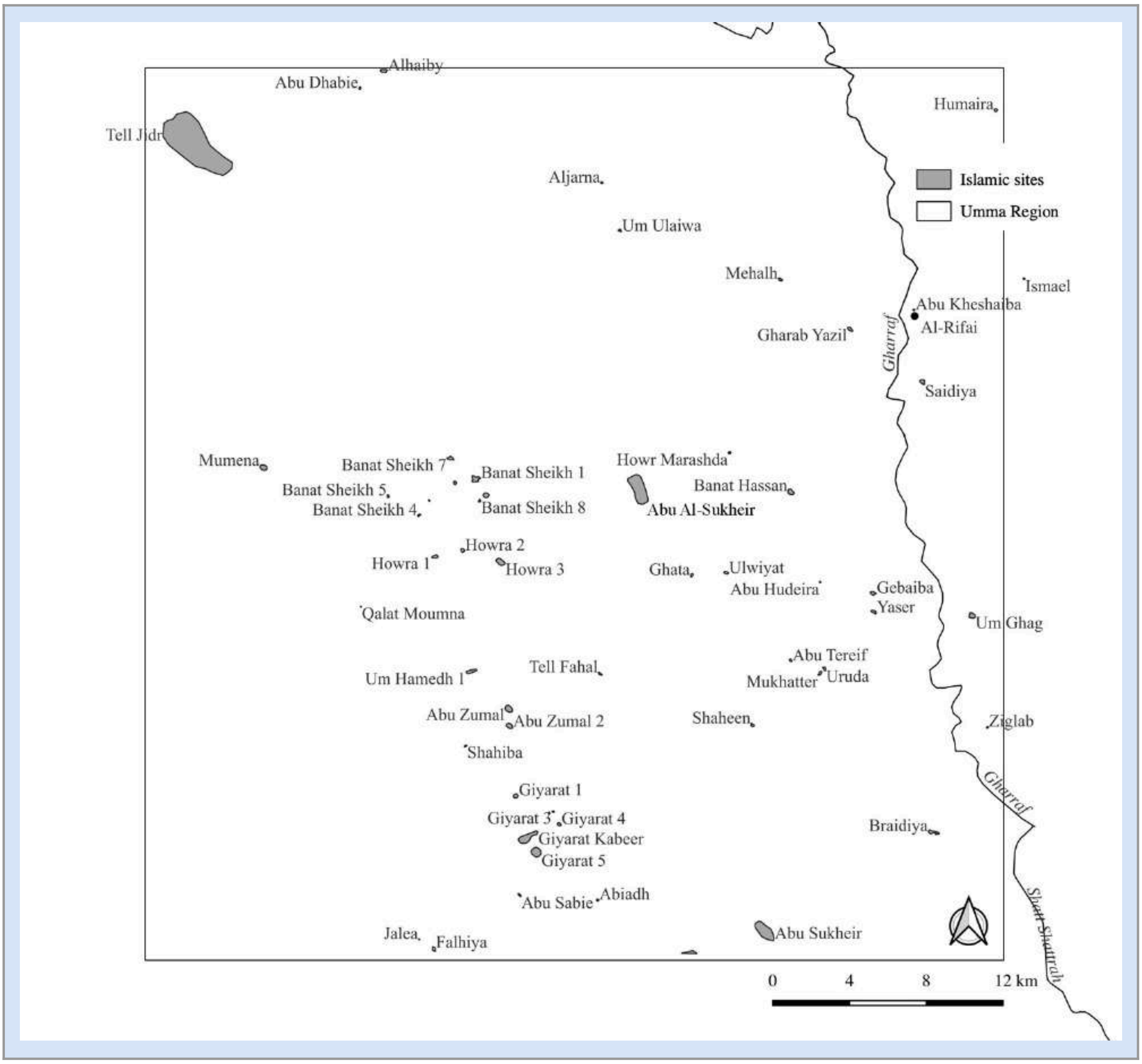

Figure 17: The Islamic Period (AD 638–1924) occupation of the Umma region.

Such trends can be confirmed for the Umma region, which is now part of the district of Wasit, named after the administrative capital Wasit (Moroney 1982, 34). The Umma region did not experience as drastic a depopulation as Adams (1981) has observed further to the west (see figure 2, 3, 17 and table 3, 4). The decline in number of settlements ranged between 10 and 20 settlements and was more pronounced between the Sasanian period and Early Islamic period (AD 638–1000). Nevertheless, there is an equally, if not more drastic reduction of the total settled area when compared to the central alluvial plain further to the west (Adams 1981, 85). The total settled area shrank by nearly 75% and large cities also completely disappeared, except for Abu Al-Sukhair (105 ha) which was occupied throughout the Islamic period and Abu Sukhair (65 ha) only in the Late Islamic period (AD 1517–1924). The decline of Jidr, which had been a local center during the Sasanian Period (AD 224–651) accounts for most of that loss. The settlement pattern is now characterized by small hamlets and villages clustered near to each other and spread all over the region. As have been mentioned earlier the region became very swampy in the middle/late Sasanian period and it is possible that many of the smaller settlements are habitations within marshland.

19. Conclusion

The Thi Qar Survey data for the Umma region yielded varied results in that they confirm but also considerably deviate from many trends observed by Adams (1981) in the settlement history of the central alluvial plain. Some of those differences might be the result of local socio-political and environmental dynamics that differed from those in other areas of southern Mesopotamia. For example, it appears that the Umma region was fairly remote in the empires of the Late Antiquity, as for example in the Achaemenid Empire (539–330 BC). While other regions in southern Mesopotamia experienced a noticeable increase in population and urbanism as well as agricultural expansion, the Umma region declined from a period of revival in the preceding Neo-Babylonian period (620–539 BC). Similarly, the expansion of agricultural land, settled area as well as an increase in the number of settlements in the Sasanian period (AD 224–651) was not as significant as it was in the central alluvial plain, possibly because the Umma region was located at the very tail end of large irrigation schemes, where the access to water was not as favorable or reliable as further upstream.

Diverging trends in the settlement history of the Umma region, however, are equally related to methodology. As mentioned earlier my survey approach was much more sensitive to the detection of smaller sites, which had often been missed in the earlier motorized surveys. For one, I visited the region much more frequently in my role as a SBHA inspector of Thi Qar province, than any of the earlier foreign surveyors. In addition, the area has in the meantime also become much more accessible for archaeological research due to a drastic decline of agricultural cultivation in the region. Furthermore, remote sensing and declassified satellite imagery have proved to be very important tools in the detection of unknown and more importantly small archaeological sites; tools that were not available in the 1950s and 1960s. The ability to detect and record smaller sites and sites of all sizes more comprehensively shed new light on the complexities of the settlement behavior in southern Mesopotamia. For one, I was able to show that the remarkable process of urbanization during the Early Dynastic period (2900–2700 BC) did not occur on the expense of rural migration and depopulation. To the contrary, the density of rural settlements increased as well, which questions earlier interpretation of a countryside rendered uninhabitable by inter-city-state warfare.

The ability to detect and record smaller sites more comprehensively also provided important insights into settlement patterns which are more closely related to the exploitation of resources other than agricultural land along natural and artificial waterways. As I have shown for several periods, settlement clusters, or settlement patterns that did not show a strict alignment along an ancient watercourse might be settlements within marshlands. The interpretations of earlier settlement data have not sufficiently considered the importance of maritime and marsh resources in ancient economies. As I was able to show, access to march and wetland resources appears to have supported Ubaid communities (5000–4000 BC) close to urban-scale dimensions (e.g. Abu Jarabie [11 ha] and Hamdha [16 ha]). Moreover, the documentation of many clusters of small settlements for nearly all historical periods, indicates the importance of wetlands in the economy of early communities and later states and empires and might even have at times been considered a place to settle, to enjoy greater independence from state institutions.

Acknowledgements

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مسح محافظة ذي قار: نتائج من مسح منطقة اوما (العراق)

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الملخص

بدأت المسوحات الأثرية الاستكشافية في العراق أواخر ثلاثينيات القرن الماضي عبر "مشروع حوض ديالى الأثري"، والذي أسس لمنهجية علمية طُبقت لاحقاً في جنوب العراق، مثل مسوحات "سومر الوسطى" ومناطق أور، ولجش، وإقليم بحيرة الحمار، والمنطقة المحيطة بمدينة كيش التاريخية. وتُعد المسوحات التي قادها روبرت آدامز بالتعاون مع باحثين آخرين الأبرز والأكثر تأثيراً في علم الآثار بلاد الرافدين؛ حيث تمكن من مسح ثلث السهل الرسوبي كاملًا على مدار عقدين (1956–1975) رغم الظروف البالغة الصعوبة وغياب الخرائط المسبقة للمنطقة. واجهت هذه الجهود تحديات طبيعية وبيئية وعرة منعت تطبيق عينات منتظمة، مثل المستنقعات، والبحيرات، والكثبان الرملية، وشبكات القنوات الزراعية المعقدة. أدت هذه العقبات، إلى جانب الكثافة الزراعية في بعض المناطق كشرق محافظة ذي قار، إلى بقاء مساحات شاسعة وعناصر حيوية دون استكشاف؛ ومنها المناطق الممتدة من مدينة أمّا الأثرية نحو نهر شط الغراف، والسهول الواقعة بين جيرسو ولجش، والمناطق الصحراوية والأهوار الجنوبية المحيطة بأور وإريدو. ورغم أن الدراسات الحديثة القائمة على الاستشعار عن بعد وتحليل صور الأقمار الصناعية (مثل صور كورونا) قد أكدت وجود مواقع أثرية عديدة غفلت عنها المسوحات القديمة، إلا أن التقييمات الميدانية الحديثة لا تزال محدودة وضئيلة حتى الآن. أظهرت بيانات مسح ذي قار لمنطقة «أوُما» (Umma) نتائج متباينة؛ حيث إنها تؤكد بعض التوجهات السائدة وتختلف اختلافاً كبيراً عن توجهات أخرى رصدها (آدامز، 1981) في التاريخ الاستيطاني للسهل الرسوبي الأوسط. وقد تعزى بعض هذه الاختلافات إلى الديناميكيات الاجتماعية-السياسية والبيئية المحلية التي تميزت بها هذه المنطقة عن سائر مناطق جنوب بلاد الرافدين. على سبيل المثال، يبدو أن منطقة «أوُما» كانت نائية نسبياً إبان عهد الإمبراطوريات في العصور القديمة المتأخرة، كما كان الحال في العهد الأخميني (539–330 قبل الميلاد). فبينما شهدت مناطق أخرى في جنوب بلاد الرافدين نمواً ملحوظاً في الكثافة السكانية والتوسع الحضري والزراعي، عانت منطقة «أوُما» من تراجع بعد فترة الانتعاش التي عاشتها في العهد البابلي الحديث السابق (620–539 قبل الميلاد). وبالمثل، لم يكن التوسع في الأراضي الزراعية والمناطق المأهولة، ولا الزيادة في عدد المستوطنات خلال العهد الساساني (224–651 ميلادي)، بذات الأهمية والزخم اللذين شهدهما السهل الرسوبي الأوسط؛ ويعود ذلك غالباً إلى وقوع منطقة «أوُما» عند النهايات القصوى لمشاريع الري الكبرى، حيث لم يكن الوصول إلى مصادر المياه يسيراً أو مضموناً كما هو الحال في أعالي الأنهار.